

# The identification of extensive samples of motor units in human muscles reveals diverse effects of neuromodulatory inputs on the rate coding

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## eLife Assessment

Leveraging state-of-the-art experimental and analytical approaches, this **important** study characterizes the recruitment and activation of large populations of human motor units during slow isometric contractions in two lower limb muscles. Evidence for the main claims is **solid** and advances our understanding of how humans generate and control voluntary force.

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## Abstract

Movements are performed by motoneurons transforming synaptic inputs into an activation signal that controls muscle force. The control signal emerges from interactions between ionotropic and neuromodulatory inputs to motoneurons. Critically, these interactions vary across motoneuron pools and differ between muscles. To provide the most comprehensive framework to date of motor unit activity during isometric contractions, we identified the firing activity of extensive samples of motor units in the Tibialis Anterior ( $129 \pm 44$  per participant;  $n=8$ ) and the Vastus Lateralis ( $130 \pm 63$  per participant;  $n=8$ ) muscles during isometric contractions of up to 80% of maximal force. From this unique dataset, the rate coding of each motor unit was characterised as the relation between its instantaneous firing rate and the applied force, with the assumption that the linear increase in isometric force reflects a proportional increase in the net synaptic excitatory inputs received by the motoneuron. This relation was characterised with a natural logarithm function that comprised two stages. The initial stage was marked by a steep acceleration of firing rate, which was greater for low- than medium- and high-threshold motor units. The second stage comprised a linear increase in firing rate, which was greater for high- than medium- and low-threshold motor units. Changes in firing rate were largely non-linear during the ramp-up and ramp-down phases of the task, but with significant prolonged firing activity only evident for

medium-threshold motor units. Contrary to what is usually assumed, our results demonstrate that the firing rate of each motor unit can follow a large variety of trends with force across the pool. From a neural control perspective, these findings indicate how motor unit pools use gain control to transform inputs with limited bandwidths into an intended muscle force.

## Introduction

Human muscles are versatile effectors producing forces that span several orders of magnitude. They allow humans to perform a broad range of motor tasks with the same limbs, such as a surgeon closing an incision or a climber who grasps supports on a cliff. This versatility relies on a unique structure that converts neural inputs into muscle force; that is, the motor unit, which comprises an alpha motoneuron and the muscle fibres it innervates (Sherrington, 1925 [\[1\]](#); Heckman and Enoka, 2012 [\[2\]](#)). The nervous system controls muscle force by modulating the number of active motor units (recruitment) and their firing rates [rate coding; (Enoka and Duchateau, 2017 [\[3\]](#))]. This control strategy involves projecting a substantial level of common excitatory synaptic inputs to motoneurons (Farina et al., 2014a [\[4\]](#); Farina and Negro, 2015 [\[5\]](#); Negro et al., 2016a [\[6\]](#)) that recruits them in a fixed order following the size principle (Henneman, 1957 [\[7\]](#)).

Although we have a clear understanding of the general strategies used to control muscle force, we still lack a full picture of the detailed organisation of the firing activities of motor units. One of the key challenges in the field is the ability to identify the concurrent firing activity of many motor units spanning the range of recruitment thresholds observed in human muscles (Farina and Holobar, 2016 [\[8\]](#)). Indeed, much of our knowledge on the modulation of motor unit activity still derives from animal preparations (Hounsgaard et al., 1988 [\[9\]](#); Bennett et al., 1998b; Lee and Heckman, 2000 [\[10\]](#)) or from serial recordings of a few concurrent motor units in humans (Desmedt and Godaux, 1977b [\[11\]](#); Bigland-Ritchie et al., 1983 [\[12\]](#); Oya et al., 2009 [\[13\]](#); Kirk et al., 2016 [\[14\]](#)), mainly identified over narrow force ranges (Monster and Chan, 1980 [\[15\]](#); Fuglevand et al., 2015 [\[16\]](#); Johnson et al., 2017 [\[17\]](#); Revill and Fuglevand, 2017 [\[18\]](#)).

Animal studies have shown two stages in the rate coding of individual motor units to increase muscle force (amplification and rate limiting) in response to linear increases in the net excitatory synaptic inputs (Bennett et al., 1998a; Lee and Heckman, 2000 [\[10\]](#); Powers and Binder, 2001 [\[19\]](#); Binder et al., 2020 [\[20\]](#)). These stages represent the different responses of motoneurons to ionotropic inputs due to modulation of their intrinsic properties by metabotropic inputs acting on the somato-dendritic surfaces of motoneurons through several ion channels (Powers and Binder, 2001 [\[19\]](#); Heckman and Enoka, 2012 [\[2\]](#); Binder et al., 2020 [\[20\]](#)). Similar characteristics in firing activity have been observed in humans (Fuglevand et al., 2015 [\[16\]](#); Revill and Fuglevand, 2017 [\[18\]](#); Beauchamp et al., 2023 [\[21\]](#)), although these observations have been limited to small samples of motor units (typically < 40) with firing activities experimentally decoded over a narrow range of submaximal forces [typically <30% maximal voluntary contraction (MVC) force; (Fuglevand et al., 2015 [\[16\]](#); Revill and Fuglevand, 2017 [\[18\]](#); Beauchamp et al., 2023 [\[21\]](#))]. It is difficult to infer from these studies a control scheme that is generalisable to an entire motor unit pool and to all contraction levels (De Luca, 1985 [\[22\]](#)).

The combination of arrays of electromyographic (EMG) electrodes with modern source-separation algorithms set the path for the identification of populations of active motor units - and their motoneurons - in humans (Farina and Holobar, 2016 [\[8\]](#); Negro et al., 2016b [\[6\]](#)). Recent studies have identified the concurrent firing activity of up to 60 motor units per contraction and per muscle by increasing the density of electrodes (Muceli et al., 2022 [\[23\]](#); Caillet et al., 2023b [\[24\]](#)). Here, we further extended this approach by tracking motor units across contractions at target forces that ranged from 10 to 80 %MVC, based on the unique spatial distribution of motor unit action potentials across the array of surface electrodes (Farina et al., 2008 [\[25\]](#); Martinez-Valdes et al., 2017 [\[26\]](#)). We

were able to identify up to ~200 unique active motor units per muscle and per participant in two human muscles *in vivo*, yielding extensive samples of motor units that are representative of the motoneuron pools (Caillet et al., 2023a). With this approach, we described the non-linear transformation of the net excitatory synaptic inputs into firing activity for individual motoneurons (Heckman and Enoka, 2012; Binder et al., 2020). We specifically focused on the non-linearities between the changes in firing rate and force: amplification, saturation, and hysteresis (Hounsgaard et al., 1988; Powers and Heckman, 2017; Binder et al., 2020). The results provided a framework picture of the rate coding of motor units during large and slow increases and decreases in an applied isometric force.

Having access to the rate coding of motoneuron pools allowed us to understand how ionotropic and neuromodulatory inputs combine to modulate force, confirming some details of hypothetical force-control schemes in humans. For example, experimental (Wei et al., 2014; Naufel et al., 2019) and computational (Powers and Heckman, 2017) studies have proposed a gain-control mechanism driven by neuromodulatory inputs to motoneurons. One common conclusion of these studies is that pools of motor units involved in fine motor tasks have a low gain, whereas additional motor units activated during more forceful tasks exhibit a progressively increase in gain (Wei et al., 2014; Powers and Heckman, 2017; Naufel et al., 2019; Binder et al., 2020). However, this conclusion is based on results from motor units in animals (Powers and Binder, 2001) and humans (Wei et al., 2014; Goodlich et al., 2022; Henderson et al., 2022) while experimentally modulating inputs, or by fitting a non-linear model predicting muscle activation from excitatory synaptic inputs at multiple contraction levels (Naufel et al., 2019). In contrast, we addressed this question by identifying the concurrent activity of hundreds of motor units spanning most of the range of recruitment thresholds *in vivo* in humans.

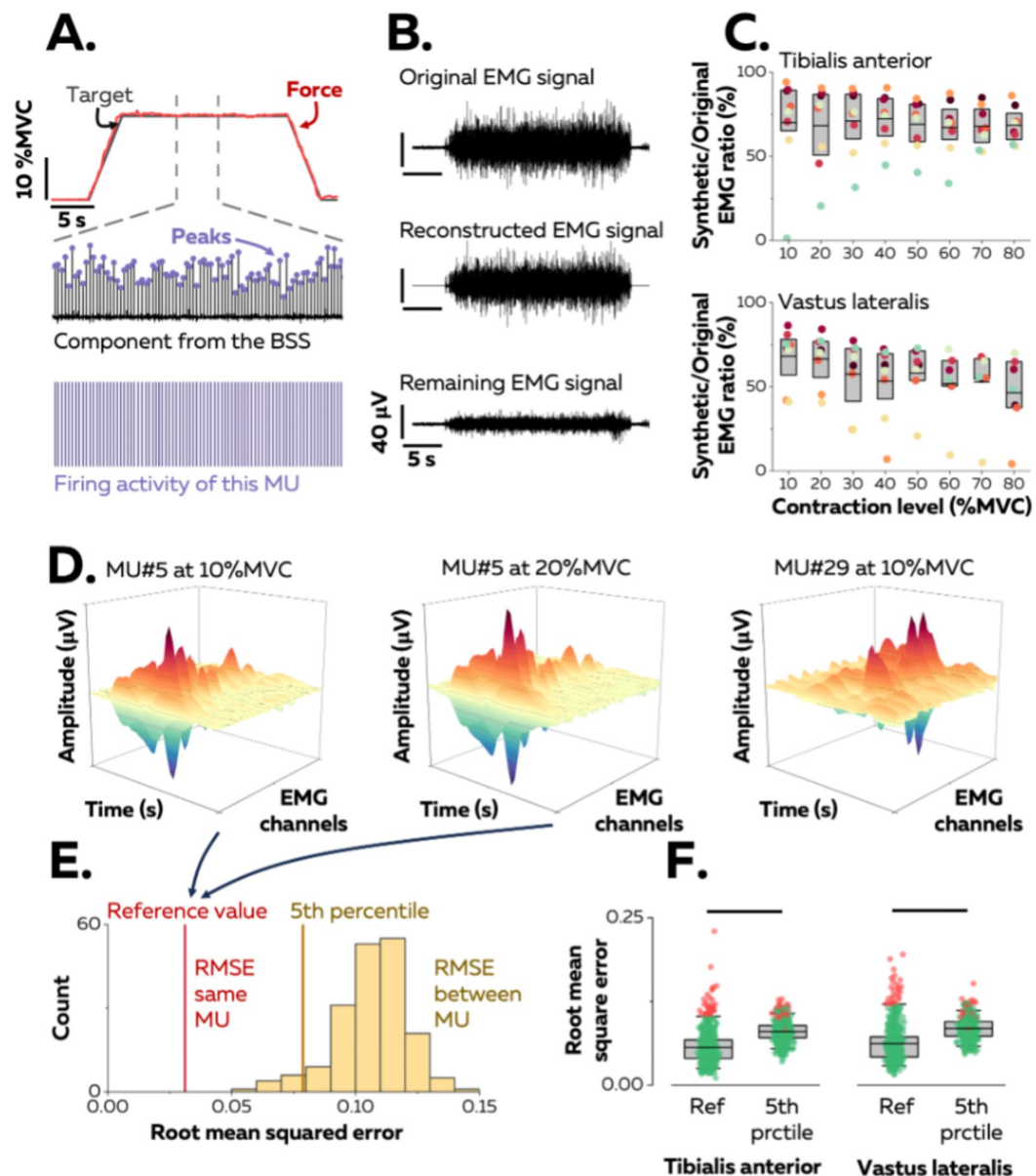
## Results

### Identification and tracking of individual motor units

16 participants performed either isometric dorsiflexion ( $n = 8$ ) or knee extension tasks ( $n = 8$ ) while we recorded the EMG activity of the tibialis anterior (TA - dorsiflexion) or the vastus lateralis (VL - knee extension) with four arrays of 64 surface electrodes (256 electrodes per muscle). The motoneuron pools of these two muscles of the lower limb receive a large part of common input (Laine et al., 2015; Negro et al., 2016a), constraining the recruitment of their motor units in a fixed order across tasks. They are therefore good candidates for an accurate description of rate coding. Moreover, we wanted to determine whether differences in rate coding observed between proximal and distal muscles in the upper limb (De Luca et al., 1982) were also present in the lower limb.

The experimental tasks comprised isometric contractions with a ramp-up, a plateau, and a ramp-down. The ramp-up and ramp-down phases were performed slowly compared with contractile speeds observed during activities of daily living. They were performed at a constant rate of 5% MVC.s<sup>-1</sup> and the force plateau was maintained for either 10-s (70-80 % MVC), 15-s (50-60% MVC), or 20-s (10-40 % MVC) (Figure 1A). The target during the plateau ranged from 10 to 80 %MVC in increments of 10 %MVC (randomised order).

A source-separation algorithm (Farina and Holobar, 2016; Negro et al., 2016b) was applied to the EMG signals to extract motor unit pulse trains, from which discharge times were automatically identified. All identified motor unit firings were visually inspected and manually edited when necessary (Avrillon et al., 2023). Because the signal was stationary during the plateau, it was possible to estimate reliable separation vectors for large samples of motor units with source-separation algorithms (Figure 1A; (Farina and Holobar, 2016; Negro et al., 2016b)). The average number of motor units identified in each contraction per participant was  $42 \pm 24$  (25<sup>th</sup>–



**Figure 1.**

### Identification of motor units in two human muscles.

A. We used a blind source separation (BSS) algorithm to decompose the overlapping activity of motor units (MU) into spike trains during a force-matching trapezoidal task (red trace). B. We reconstructed synthetic EMG signals by summing the trains of action potentials from all the identified motor units and interpreting the remaining EMG signal as the part of the signal not explained by the decomposition. C. We calculated the ratio between the powers of synthetic and original EMG signals to estimate the proportion of the signal variance explained by the decomposition. Each data point indicates the average value for one participant. D. We estimated the uniqueness of each identified motor unit within the pool by calculating the root-mean-square error (RMSE) between the distributions of action potentials of the same motor unit across contractions (two panels on the left, reference value in E), and between motor units (left vs. right panels, distribution of RMSE between motor units in yellow in E). F. Each motor unit was unique within the pool when the RMSE between its distributions of action potentials across target forces (reference value) was less than the 5<sup>th</sup> percentile of the distribution of RMSE with the rest of the motor units. Motor units considered as outliers in F (red data points) were removed from the analysis due to potential errors in tracking between contractions. Each data point is a motor unit, the box represents 25<sup>th</sup>-75<sup>th</sup> percentiles of the distribution of data, and the black line shows the median. The horizontal thick line denotes a statistical difference between reference values and 5<sup>th</sup> percentiles for each muscle.

75<sup>th</sup> percentile: 24 – 53, up to 95; [Table 1](#)) motor units from the TA and  $33 \pm 15$  (25<sup>th</sup>–75<sup>th</sup> percentile: 23 – 47, up to 71; [Table 1](#)) motor units from the VL. The datasets from all target forces were merged into a sample of unique motor units per muscle and participant that spanned most of the operating range of recruitment thresholds observed in humans (1<sup>st</sup>–99<sup>th</sup> percentile: 0.9–73.4% MVC; see below).

The proportion of the EMG signal represented by the identified motor units was estimated by reconstructing a synthetic EMG signal from the firing activity. To do so, the discharge times were used as triggers to segment the differentiated EMG signals over a window of 25 ms that yielded averaged action potential waveforms for each motor unit ([Figure 1A](#)). The action potentials were then convolved with the discharge times to obtain trains of action potentials, and all the trains of the identified motor units were summed to reconstruct the synthetic EMG signal. The ratio between the powers of synthetic and experimental EMG signals was calculated ([Figure 1B](#)): it was  $69.3 \pm 17.3\%$  (25<sup>th</sup>–75<sup>th</sup> percentile: 59.3 – 83.6%, up to 94.2%) for TA and  $55.2 \pm 19.5\%$  (25<sup>th</sup>–75<sup>th</sup> percentile: 50.0 – 71.9%, up to 86.5%) for VL ([Figure 1C](#)). These values indicate that most of the recorded surface EMG signals were successfully decomposed into motor unit activity.

Motor units were tracked between contractions using their unique spatial distribution of action potentials ([Figure 1D](#); (Martinez-Valdes et al., 2017)). This method was validated by using both simulated EMG signals and two-source validation with simultaneous recordings of firing activity from intramuscular and surface EMG signals ([Figure S1](#)). On average, we tracked  $67.1 \pm 10.0\%$  (25<sup>th</sup>–75<sup>th</sup> percentile: 53.9 – 80.1%) of the motor units between consecutive target forces (10% increments, e.g., between 10% and 20% MVC) for TA and  $57.2 \pm 5.1\%$  (25<sup>th</sup>–75<sup>th</sup> percentile: 46.6 – 68.3%) of the motor units for VL ([Figure S2](#)). There are two explanations for the inability to track all motor units across consecutive target forces: i) some motor units are recruited at higher targets only; ii) it is challenging to track small motor units beyond a few target forces due to a lower signal-to-noise ratio when larger motor units are recruited, or signal cancellation (Keenan et al., 2005; Farina et al., 2014b). In total, we identified  $129 \pm 44$  motor units (25<sup>th</sup>–75<sup>th</sup> percentile: 100 – 164, up to 194) per participant in TA and  $130 \pm 63$  motor units (25<sup>th</sup>–75<sup>th</sup> percentile: 103 – 173, up to 199) per participant in VL.

The accuracy of the tracking method was further tested by confirming the uniqueness of each motor unit within the identified sample, assuming that each motor unit had a unique representation across the array of surface electrodes (Farina et al., 2008). This was accomplished by calculating the root mean-square error (RMSE) between their action potentials across the electrodes relative to those of the rest of the motor units ([Figure 1E](#)). The RMSE between the action potentials of the same motor unit tracked across contractions was calculated as a reference and compared with the 5<sup>th</sup> percentile of the distribution of RMSE between motor units ([Figure 1F](#)). The reference value was typically less than the 5<sup>th</sup> percentile of the RMSE calculated with the action potentials of the other motor units (TA:  $5.6 \pm 2.4$  vs.  $8.0 \pm 1.5\%$ ;  $p < 0.001$ ; VL:  $6.2 \pm 2.8\%$  vs.  $8.5 \pm 1.6\%$ ;  $p < 0.001$ ). Inspection of the data for each motor unit revealed that 92.1% (TA) and 87.0% (VL) of the motor units had a lower RMSE between target forces compared with the other motor units (<5<sup>th</sup> percentile). Of note, motor units with the highest reference values (>95<sup>th</sup> percentile) were considered as outliers due to tracking errors and were excluded from the subsequent analyses ([Figure 1F](#)). We excluded 34 motor units from TA and 28 from VL.

## Non-linear rate coding during the ramp-up phase – amplification and rate limiting

The input-output function for each motoneuron was characterised as the relation between its instantaneous firing rate and the applied force during the ramp-up phase of the contractions ([Figure 2A](#)). The linear increase in force was assumed to reflect a proportional increase in the net synaptic excitatory inputs received by the motoneurons, as proposed previously (Fuglevand et al., 1993; Revill and Fuglevand, 2017). Thus, any deviation from a linear increase in firing rate

|           | 10%              | 20%              | 30%              | 40%              | 50%              | 60%              | 70%              | 80%             |
|-----------|------------------|------------------|------------------|------------------|------------------|------------------|------------------|-----------------|
| <i>TA</i> | 38±25<br>(1-73)  | 45±24<br>(21-83) | 50±27<br>(24-95) | 49±26<br>(24-93) | 45±27<br>(15-93) | 40±25<br>(14-80) | 37±26<br>(14-84) | 34±19<br>(9-60) |
| <i>VL</i> | 44±17<br>(17-63) | 46±20<br>(19-71) | 42±20<br>(10-67) | 34±18<br>(3-56)  | 32±15<br>(5-53)  | 26±13<br>(3-38)  | 26±14<br>(1-38)  | 20±13<br>(1-37) |

TA, tibialis anterior, VL, vastus lateralis

**Table 1**

**Mean ± standard deviation (range) for the number of motor units across the eight target forces and two muscles**



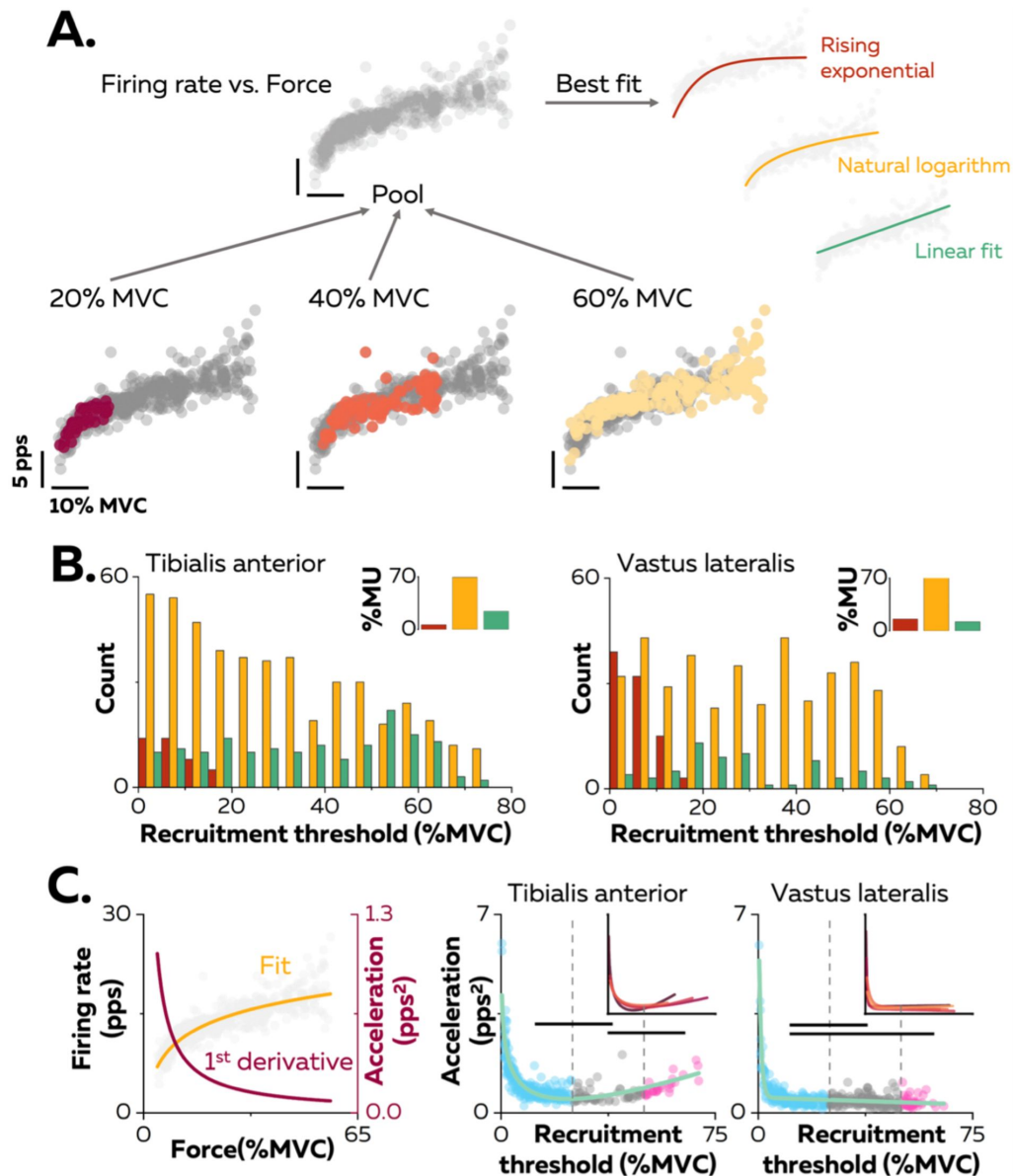
with respect to a linear increase in force presumably reflects the influence of neuromodulatory inputs on motoneuron gain (Johnson et al., 2017 [↗](#); Revill and Fuglevand, 2017 [↗](#); Beauchamp et al., 2023 [↗](#)), or a saturation of the motoneuron firing rate (Fuglevand et al., 2015 [↗](#); Revill and Fuglevand, 2017 [↗](#)).

The association between the firing rate and the force was characterised by comparing three curve fits: i) linear - an increase in firing rate proportional to the increase in the net synaptic excitatory input, ii) rising exponential - an initial acceleration of firing rates followed by full saturation (De Luca and Contessa, 2012 [↗](#); Fuglevand et al., 2015 [↗](#); Revill and Fuglevand, 2017 [↗](#)), and iii) natural logarithm - an initial acceleration of firing rate followed by a slower constant increase in firing rate that reflects a rate limiting effect [Figure 2A [↗](#); (Bennett et al., 1998a; Lee and Heckman, 2000 [↗](#); Powers et al., 2012 [↗](#))]. Of note, this analysis was performed on each unique motor unit with the data pooled across all contractions (Figure 2A [↗](#)). The result was that 69.5% of the motor units in TA and 72.3% of those in VL had an input-output function better fitted by a natural logarithm (Figure 2B [↗](#), inset panels).

The input-output functions of the motor units were then compared based on the distribution of recruitment thresholds (1<sup>st</sup>–99<sup>th</sup> percentile: 0.9–73.4% MVC). Motor units were clustered into three groups with equal ranges of recruitment thresholds: low-(0 to 25% MVC), medium-(25 to 50% MVC), and high-(50 to 75% MVC) threshold motor units (Figure 2B [↗](#)). A significant percentage of input-output functions for low threshold motor units was better fitted with a rising exponential function (12.5% for the TA, 30.9% for the VL), highlighting a steep acceleration of firing rate followed by a full saturation. This pattern was not observed in medium- and high-threshold motor units (0% for TA and VL). In contrast, a significant percentage of input-output functions of high threshold motoneurons was better fitted with a linear function for TA (39.6%), although the input-output functions of high threshold motor units from VL were still better fitted with natural logarithms (87.9%) than linear functions (12.1%).

We further described the first stage of rate coding by estimating the acceleration of firing rate (Figure 2C [↗](#); Figure S3 [↗](#)). The initial acceleration of the firing rate, likely due to the amplification of synaptic inputs by persistent inward currents (Lee and Heckman, 2000 [↗](#)), was calculated as the value of the first derivative of the natural logarithm function at the recruitment threshold (Figure 2C [↗](#)). These values were compared between motor units using a linear mixed effect model, with *Muscle* (TA; VL) and *Threshold* (low; medium; high) as fixed effects and *Participant* as a random effect. There was a significant effect of *Muscle* ( $F=30.4$ ;  $p<0.001$ ), *Threshold* ( $F=24.7$ ;  $p<0.001$ ), and a significant interaction *Muscle*  $\times$  *Threshold* ( $F=4.1$ ;  $p=0.017$ ). The initial acceleration of firing rate in TA was greater for the low- $(1.0\pm0.7 \text{ pps}^2$ ;  $p<0.001$ ) and the high- $(1.0\pm0.3 \text{ pps}^2$ ;  $p=0.032$ ) threshold motor units than for medium-threshold motor units ( $0.6\pm0.3 \text{ pps}^2$ ), with no difference between the low- and high-threshold motor units ( $p=0.999$ ; Figure 2C [↗](#)). Similarly, the initial acceleration of firing rate was greater for low-threshold motor units ( $0.6\pm0.6 \text{ pps}^2$ ) than for medium- $(0.4\pm0.2 \text{ pps}^2$ ;  $p<0.001$ ) and high-threshold motor units ( $0.4\pm0.2 \text{ pps}^2$ ;  $p=0.013$ ) in VL, with no difference between the medium- and high-threshold motor units ( $p=0.990$ ; Figure 2C [↗](#)). The initial acceleration of firing rate was greater for the low- and high-threshold motor units in TA compared with VL ( $p<0.001$  for both).

We also tested whether the distribution of initial accelerations of firing rate observed across the entire motor unit pool was generalisable to all the participants. Participants with either fewer than 20 motor units or with no motor units recruited below 5%MVC were excluded from this analysis (TA:  $n=4$ ; VL:  $n=3$ ). The function representing the relation between recruitment thresholds and initial acceleration were accurately fitted using the non-linear least-square fitting method (TA; adjusted R-squared: 0.77; VL; adjusted R-squared: 0.80; green lines on Figure 2C [↗](#)). We then fitted the same non-linear functions for each participant; the adjusted R-squared values increased to  $0.86 \pm 0.04$  for the TA and  $0.84 \pm 0.11$  for the VL (inset panels in Figure 2C [↗](#)).



**Figure 2.**

### Non-linear rate coding of motor units.

A. The relation between firing rate (pulses per second, pps) and the applied force during the ramp-up phase of the contraction was determined by concatenating the instantaneous firing rates for each motor unit (grey data points) recorded over all the contractions where it was identified, as shown here for one motor unit (coloured data points for contractions at 20, 40, and 60%MVC). The derived relations were then fitted with three different functions: linear (green), rising exponential (dark red), and natural logarithm (yellow), to characterise the input-output function of each motoneuron. B. The motor units were grouped according to their best fit. The graphs show the distribution of these groups as a function of recruitment thresholds (RT) for each muscle. The inset panels depict the percentage of motor units (MU) in each group. C. The initial acceleration of firing rate was derived from force-firing rate relation fitted with the natural logarithm ( $f(\text{force}) = a \cdot \ln(\text{force}) + b$ ; yellow trace) and its first derivative ( $f(\text{force}) = a/\text{force}$ ; dark red trace). The right panels show the distribution of initial acceleration values relative to recruitment threshold (RT) for all participants ( $n = 328$  motor units for TA and  $n = 393$  motor units for VL). Each data point indicates a motor unit. The horizontal thick lines denote a statistical difference between the motor units groups (low-threshold = Blue; medium-threshold = Grey; high-threshold = Pink). The green line depicts the non-linear fits of these relations for the Tibialis Anterior and the Vastus Lateralis. Similar fits were observed for all the participants (inset panels).



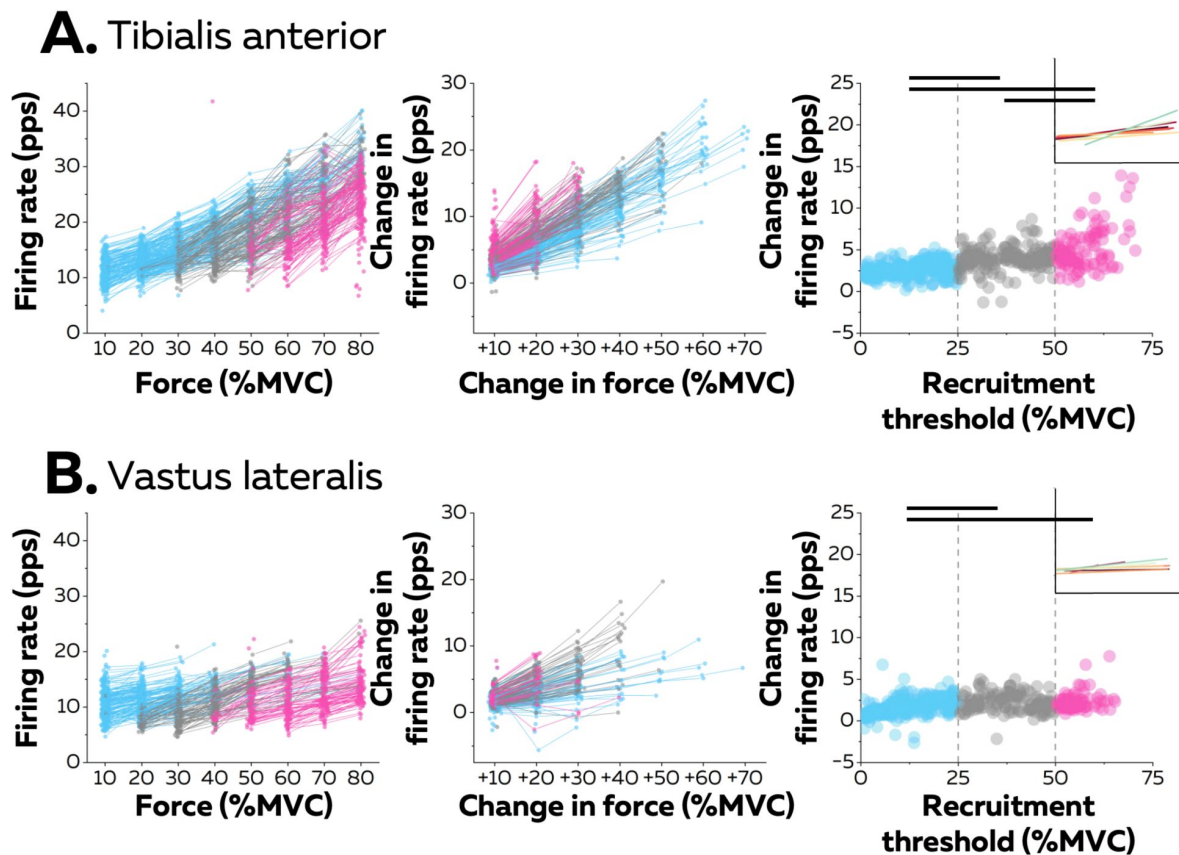
The second stage of rate coding involved a linear increase in firing rate after the initial acceleration phase. This stage was analysed using the average firing rate during the successive force plateaus for each tracked motor unit (**Figure 3**). Specifically, the increase in firing rate between consecutive contraction levels (i.e., change in force equal to 10% MVC) was compared with a linear mixed effect model with *Muscle* (TA; VL) and *Threshold* (low; medium; high) as fixed effects and *Participant* as a random effect. There were significant effects for *Muscle* ( $F=39.7$ ;  $p<0.001$ ) and *Threshold* ( $F=145.3$ ;  $p<0.001$ ), and a significant interaction *Muscle*  $\times$  *Threshold* ( $F=37.7$ ;  $p<0.001$ ). The rate of increase was significantly greater for TA ( $3.5 \pm 1.7$  pps; 25<sup>th</sup>-75<sup>th</sup> percentile: 2.4 – 4.1 pps) than for VL ( $2.0 \pm 1.1$  pps; 25<sup>th</sup>-75<sup>th</sup> percentile: 1.3 – 2.6 pps,  $p<0.001$ ). The rate of increase was significantly greater in TA for high-threshold than for low- ( $p<0.001$ ) and medium- ( $p<0.001$ ) threshold motor units and for medium- than for low-threshold motor units ( $p<0.001$ ). In contrast, the rate of increase in VL was greater for high- ( $p<0.001$ ) and medium- ( $p<0.001$ ) threshold motor units than for low-threshold motor units, with no difference between high- and medium-threshold motor units in VL ( $p=0.993$ ).

The linear functions for each participant are shown in **Figure 3**. There was a significant positive association between the rate of increase in firing rate and the recruitment threshold for all the participants (Pearson's  $r$ : TA:  $0.71 \pm 0.14$  pps and VL:  $0.57 \pm 0.22$  pps; all  $p<0.022$ ), except in one participant for each muscle (TA:  $r=0.17$ ;  $p=0.272$ ; VL:  $r=0.12$ ;  $p=0.411$ ). Most motor units (92.0% in TA and 80.9% in VL) increased firing rates by more than one pulse per second between contractions up to the maximal force tested or until tracking was not possible.

## Differences between ramp-up and ramp-down - hysteresis

Neuromodulatory inputs can prolong excitatory synaptic inputs and produce unequal recruitment and derecruitment thresholds; that is, a hysteresis. The left columns in **Figure 4** show the relation between recruitment and derecruitment thresholds within the motor unit pools (**Figure 4**). The relation for most participants was better characterised with a non-linear than a linear function (TA: 6 out of 8 participants, adjusted R-squared  $>0.92$ ; VL: 7 out of 8 participants, adjusted R-squared  $>0.83$ ). The difference between recruitment and derecruitment thresholds was compared with a linear mixed effect model with *Muscle* (TA; VL) and *Threshold* (low; medium; high) as fixed effects and *Participant* as a random effect. There was a significant effect of *Threshold* ( $F=83.0$ ;  $p<0.001$ ) and a significant interaction *Muscle*  $\times$  *Threshold* ( $F=30.1$ ;  $p<0.001$ ), but no effect of *Muscle* ( $F=2.5$ ;  $p=0.137$ ). We further tested for each group of motor units whether the difference between the two thresholds was significantly different from 0 (absence of hysteresis). The result was that only medium-threshold motor units had a significant positive hysteresis (derecruitment threshold was lower) in both muscles (TA:  $-3.8 \pm 7.2$  pps;  $p=0.013$ ; VL:  $-3.1 \pm 6.0$  pps;  $p=0.026$ ), whereas low-threshold motor units from both muscles (TA:  $-1.5 \pm 4.8$ ; VL:  $-1.0 \pm 3.2$  pps;  $p>0.221$ ) did not exhibit significant hysteresis. Although high-threshold motor units from TA followed the same trend ( $-1.7 \pm 6.0$  pps;  $p<0.437$ ), high-threshold motor units from VL exhibited a negative hysteresis (VL:  $+3.2 \pm 6.7$  pps;  $p=0.039$ ), which indicated that derecruitment threshold was greater than the recruitment threshold.

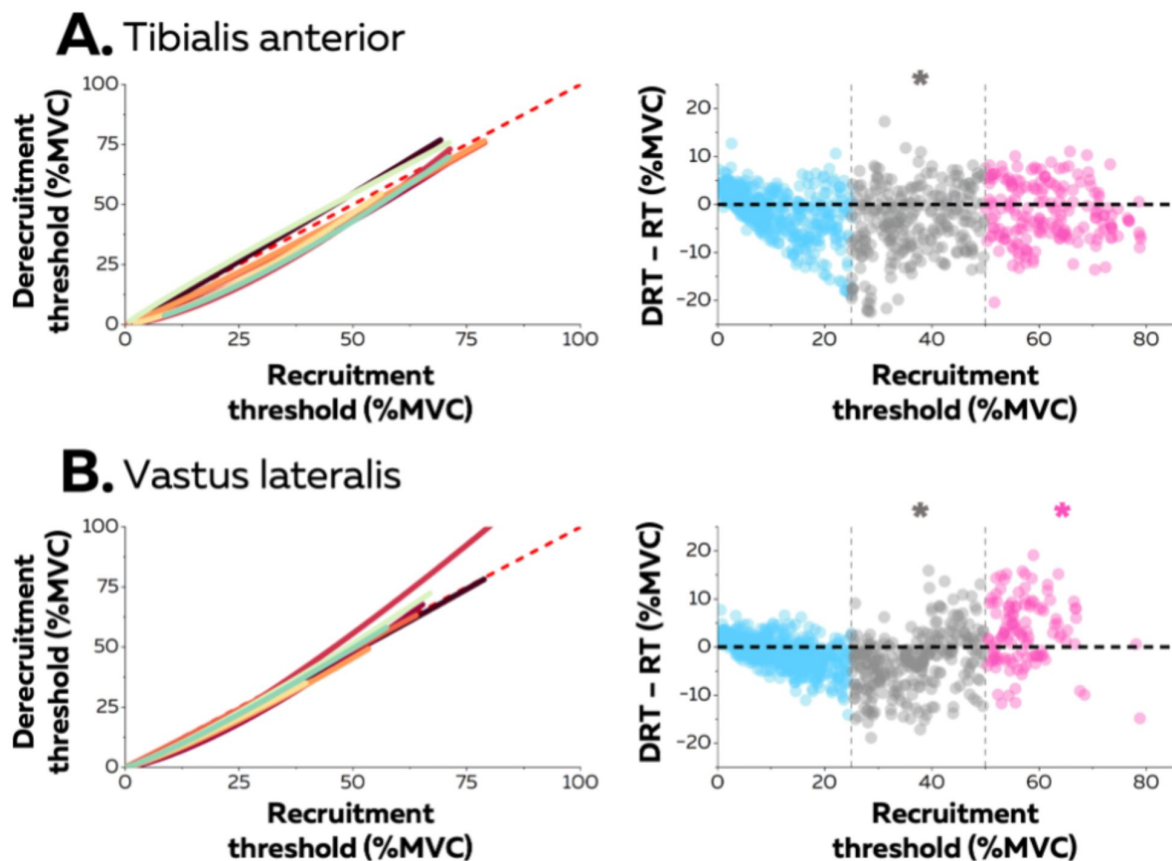
We also compared the type of the best fit between motor unit firing rate and force between the ramp-up and ramp-down phases of the isometric contraction. Most motor units (TA: 60.2%; VL: 76.7%) were best fit with the same relation during the two phases. Although the percentage of associations better fitted with linear functions increased for the ramp-down phase for the TA (from 25.4% to 41.9% of motor units), there was no change between ramps for the VL (from 12.8% to 19.4% of motor units). Thus, most motor units kept a non-linear relation during the ramp-down phase.



**Figure 3.**

#### Motor unit firing rates across contraction levels.

The left column shows average firing rate (pulses per second, pps) during the force plateaus for each tracked motor unit across contraction levels for all participants from TA ( $n = 998$  motor units; A) and VL ( $n = 1016$  motor units; B). Each data point indicates one motor unit, and each line connects the firing rates of this motor unit across contractions. The colour scale identifies the three groups of motor units based on recruitment threshold: low (blue), medium (grey), and high (pink). The middle column depicts the change in firing rates between contractions separated by 10% to 70% of the maximal voluntary contraction level (MVC) of force. The right column shows the relation between the rate of increase in firing rate between successive target forces (e.g., between 10% and 20%MVC) and the recruitment threshold of the motor unit. These relations were fitted with a linear function (coloured lines in the inset panels) for each participant. The horizontal thick line denotes a statistical difference between motor units grouped by recruitment thresholds.



**Figure 4.**

#### Hysteresis between recruitment and derecruitment thresholds.

The left column depicts the relations between the recruitment and derecruitment thresholds of each motor unit from TA (A) and VL (B). These relations were fitted for each participant (coloured lines) using either non-linear or linear regressions. The values below the dashed red line (recruitment threshold = derecruitment threshold) show a positive hysteresis between recruitment and derecruitment thresholds, the values above a negative hysteresis. The right column shows the difference between the recruitment and derecruitment thresholds, with negative values showing a positive hysteresis with recruitment threshold greater than derecruitment threshold and the positive values indicating the converse (a negative hysteresis). Each data point is a motor unit. The asterisk denotes a statistical difference between the hysteresis values for motor units grouped according to recruitment threshold (low = Blue; medium = Grey; high = Pink) and the absence of a hysteresis (dashed horizontal line).

## Discussion

No previous studies have identified to date the firing activity of >100 motor units spanning most of the spectrum of recruitment thresholds from the same individual, whether in animals or humans. According to previous estimates, the number of motor units detectable with surface EMG in human would be  $200 \pm 61$  for TA and  $146 \pm 29$  for VL (Duchateau and Enoka, 2022). We have therefore identified in this study most of the active motor units present in the recorded volume for each muscle (**Figure 1**), which has allowed us to infer the rate coding of pools of motor units during slow isometric contractions. Overall, we found that motor units within a pool exhibit distinct rate coding with changes in force level (**Figure 2** and **3**), which contrasts with the long-held belief that rate coding is similar across motor units from the same pool (Fuglevand et al., 1993; De Luca and Contessa, 2012).

The fast initial acceleration stage reflects the amplification of synaptic inputs through the activation of persistent inward currents via voltage-gated sodium and calcium channels (Bennett et al., 1998a; Lee and Heckman, 2000; Heckman and Enoka, 2012; Binder et al., 2020). Although noradrenergic and serotonergic synapses that generate persistent inward currents are widespread within motor unit pools (Maratta et al., 2015), the influence of the neuromodulatory inputs is greater during this phase for low- than for high-threshold motor units. This is presumably due to a larger ratio between the amplitude of inward currents and the input conductance in low threshold motoneurons (Huh et al., 2017). The level of recurrent and reciprocal inhibition has also probably increased with the increase in force during the ramp up, progressively blunting the effect of persistent inward currents for late-recruited motor units (Kuo et al., 2003; Hynsgstrom et al., 2007; Revill and Fuglevand, 2017). This may also explain the larger percentage of high-threshold motor units with a linear fit for the firing rate/force relation (**Figure 2**), as the integration of larger inhibitory inputs should linearise the firing rate/force relation (Revill and Fuglevand, 2017).

The second stage of rate coding involves a slower linear increase in firing rate, as characterised by the second part of a logarithmic function (**Figures 2-3**). This stage corresponds to mechanisms previously referred to as ‘rate limiting’ (Heckman and Binder, 1993; Heckman and Enoka, 2012; Powers and Heckman, 2017) or ‘saturation’ (De Luca and Contessa, 2012; Fuglevand et al., 2015), and was more pronounced for low- than high-threshold motor units (**Figure 3**). This difference may be explained by smaller excitatory synaptic inputs onto low- than high-threshold motoneurons (Powers and Binder, 2001; Heckman and Enoka, 2012), lower synaptic driving potential of the dendritic membrane (Powers and Binder, 2000; Cushing et al., 2005; Fuglevand et al., 2015), and longer and larger afterhyperpolarisation phase in low- than high-threshold motoneurons (Bakels and Kernell, 1993; Gardiner, 1993; Deardorff et al., 2013; Caillet et al., 2022).

Taken together, these results show how ionotropic and neuromodulatory inputs to motoneurons uniquely combine to generate distinct rate coding across the pool, even if a more direct manipulation of the sources of neuromodulatory and ionotropic inputs will be required to directly estimate their interactions. While the size of the motor unit determines its recruitment threshold, neuromodulatory inputs determine its gain. Thus, in a similar fashion as size imposes a fixed recruitment order (Henneman, 1957), neuromodulatory inputs determine a spectrum of variations in motor unit firing rates without the need to differentiate the ionotropic inputs to each motoneuron. [**Figure 2C-3-S3**; (Powers and Heckman, 2017; Bracklein et al., 2022; Chardon et al., 2023; Škarabot et al., 2023)]. Because high-threshold motor units exhibit a higher gain than low-threshold motor units (**Figure 3**), their firing rate could eventually reach similar or even greater values than that of low-threshold motor units during strong contractions (Gydikov and Kosarov, 1974; Moritz et al., 2005; Oya et al., 2009; Škarabot et al., 2023).

This indicates that the onion skin principle (De Luca and Contessa, 2012; 2015) may not hold in all muscles and for all contraction forces (Gydikov and Kosarov, 1974; Moritz et al., 2005; Oya et al., 2009; Škarabot et al., 2023). In addition, rate coding patterns should also vary with the pattern of contractions, with fast contractions lowering the range of recruitment thresholds within motoneuron pools (Desmedt and Godaux, 1977a, 1979; van Bolhuis et al., 1997). The variability in rate coding observed here between motor units from the same pool could lead to small deviations from the size principle sometimes observed between pairs of units during isometric contractions with various patterns of force (Desmedt and Godaux, 1979; Marshall et al., 2022) or during the derecruitment phase (Bracklein et al., 2022).

It is also worth noting that the profile of rate coding mostly followed the same trend during the ramp-up and ramp-down phases of the contraction, but with prolonged firing activity only evident for medium-threshold motor units (**Figure 4**). This hysteresis is possible because of the bistability of the membrane potential mediated by inward currents from calcium channels, the two stable membrane states being the resting potential and a depolarised state that enables self-sustained firing (Hounsgaard et al., 1988; Lee and Heckman, 1998). The absence of statistically significant hysteresis in low threshold motoneurons may be simply explained by their early recruitment during the first percentages of force, mathematically lowering the potential amplitude of their hysteresis. Alternatively, prominent outward currents may attenuate the impact of persistent inward currents on the prolongation of their firing activity (Powers and Heckman, 2015). Similarly, the absence of hysteresis (TA) and even a negative hysteresis (VL) in high-threshold motoneurons can be explained by a briefer membrane bistability due to a faster decay of inward currents from calcium channels, and a narrower range of membrane depolarisation over which these inward currents are activated (Lee and Heckman, 1998). This result must be confirmed with a more direct proxy of the net synaptic drive, such as the firing rate of a *reference* low-threshold motor neuron used in the delta F method (Gorassini et al., 1998), or the cumulative spike train of low-threshold motor neurons (Afsharipour et al., 2020).

The increase in firing rate was also significantly greater for TA motor units than for those in VL. This difference may reflect a varying balance between excitatory/inhibitory synaptic inputs and neuromodulation due to multiple spinal circuits (Heckman and Binder, 1993; Heckman et al., 2008; Johnson et al., 2017; Powers and Heckman, 2017; Chardon et al., 2023; Škarabot et al., 2023). Specifically, the strength of recurrent and reciprocal inhibitory inputs to motoneurons innervating VL and TA, and their proportional or inverse covariation with excitatory inputs, respectively, may explain the differences in rate limiting and maximal firing rates (Heckman and Binder, 1993; Heckman et al., 2008; Johnson et al., 2017; Powers and Heckman, 2017; Chardon et al., 2023; Škarabot et al., 2023). Thus, the motor units from the VL may receive more recurrent inhibition than those of distal muscles, though direct evidence of these differences remains to be found in humans (Windhorst, 1996). Interestingly, similar differences in rate coding were previously observed between proximal and distal muscles of the upper limb (De Luca et al., 1982). However, other muscles that serve different functions within the human body, such as muscles from the face, have different rate coding characteristics with much higher firing rates (Kirk et al., 2021). Future work should investigate those muscles and other to reveal the myriads of rate coding strategies in human muscles.

Our results on rate coding characteristics of the motor unit pool provide insight on force control strategies. The accurate control of force depends on the firing activity of the population of active motor units, which effectively filters out the synaptic noise of individual motor units and primarily transforms common synaptic inputs into muscle force (Dideriksen et al., 2012; Farina et al., 2014a; Farina and Negro, 2015). Because the bandwidth of muscle force is <10 Hz during isometric contractions (Enoka and Farina, 2021), the activation of inward currents to low-threshold motoneurons at recruitment may enable them to promptly discharge at a rate (>8 pps)



that transmits the effective common synaptic inputs (<8-10 Hz) without phase distortion (Dideriksen et al., 2012 [↗](#); Farina et al., 2014a [↗](#); Farina and Negro, 2015 [↗](#)). This mechanism facilitates the accuracy of force control.

Moreover, force generation can be described as the filtering of the cumulative firing activity of active motor units with the average twitch force of their muscle units (Farina and Negro, 2015 [↗](#); Thompson et al., 2018 [↗](#)). From this perspective, at low forces, recruitment of new motor units has a higher impact on force modulation than rate coding since each recruitment impacts both the cumulative firing activity of the pool and the average twitch force. Rate coding only modulates the cumulative firing activity. Thus, the amplification of the firing rate of low-threshold motor units near their recruitment threshold instantly favours rate coding over the recruitment of additional motor units, which likely allows for smoother force control. Similarly, the progressive recruitment of motor units with a higher gain promotes faster changes in firing rates, which promotes force control accuracy across the full force range. On a different note, the steep increase in firing rate over the first percentages of the ramp-up may also enable the motor units to produce the required level of force despite having a more compliant muscle-tendon unit (Mazzo et al., 2021 [↗](#)). Overall, our results may help to design future non-linear decoders that aim to predict muscle activation or muscle force from descending inputs recorded in the motor cortex, with the will to generalise their performance across movements (Naufel et al., 2019 [↗](#)).

## Materials and Methods

### Participants

16 young individuals volunteered to participate either in the experiment on the Tibialis Anterior (n=8; age:  $27 \pm 3$ ) or on the Vastus Lateralis (n=8; age:  $27 \pm 10$ ). They had neither history of lower limb injury in the last six months before the experiments nor lower leg pain that would impair their ability to complete the experimental tasks. The study was reviewed and approved by Imperial College London (Study 18IC4685) and Comité de Protection des Personnes Ouest III (Study 23.00453.000166) and followed the standards of the declaration of Helsinki. Participants provided their informed written consent before starting the experimental session.

### Experimental setup

The two experimental sessions consisted of either a series of submaximal isometric ankle dorsiflexions or isometric knee extensions. Electromyographic (EMG) signals were recorded from either the Tibialis Anterior (TA) or the Vastus Lateralis (VL) muscles using four arrays of 64 surface electrodes for a total of 256 electrodes.

For the session of ankle dorsiflexions, participants sat on a massage table with the hips flexed at 45°, 0° being the hip neutral position, and the knees fully extended. The foot of the dominant leg (right in all participants) was fixed onto the pedal of an ankle dynamometer (OT Bioelettronica, Turin, Italy) positioned at 30° in the plantarflexion direction, 0° being the foot perpendicular to the shank. The thigh and the foot were fixed with inextensible Velcro straps. Force signals were recorded with a load cell (CCT Transducer s.a.s, Turin, Italy) connected in-series to the pedal using the same acquisition system as for the EMG recordings (EMG-Quattrocento; OT Bioelettronica, Italy).

For the session of knee extensions, participants sat on an instrumented chair with the hips flexed at 85°, 0° being the hip neutral position, and the knees flexed at 85°, 0° being the knees fully extended. The torso and the thighs were fixed to the chair with Velcro straps and the tibia were positioned against a rigid resistance connected to force sensors (Metitur, Jyväskylä, Finland). The force signals were recorded using the same acquisition system as for the EMG recordings.



The experimental session began with a warm-up that consisted of submaximal isometric contractions from 50% to 80% of the subjective MVC. Then, participants performed two maximal voluntary contractions (MVC) with two minutes of rest in between. The maximal torque was considered as the highest torque value recorded over an average window of 250 ms. The rest of the session consisted of a series of eight submaximal isometric contractions performed at 10% to 80% of the MVC with 10 %MVC increments. The pattern of the contractions followed a trapezoidal target displayed on a screen in real time with the force trace. Ramp-up and ramp-down phases were performed at a constant pace of 5 %MVC.s<sup>-1</sup>. The force plateaus were maintained for either 10-s (70-80 %MVC), 15-s (50-60 %MVC) or 20-s (10-40%MVC). The contractions were separated by 90 s of rest and their order was randomised.

## EMG recording

Surface EMG signals were recorded from the Tibialis Anterior (TA) or the Vastus Lateralis (VL) using four two-dimensional arrays of 64 electrodes (GR04MM1305 for the TA; GR08MM1305 for the VL, 13×5 gold-coated electrodes with one electrode absent on a corner; interelectrode distance: 4 and 8mm, respectively; OT Bioelettronica, Italy). The grids were positioned over the muscle bellies to cover the largest surface while staying away from the boundaries of the muscle identified by manual palpation. Before placing the electrodes, the skin was shaved and cleaned with an abrasive pad and water. A biadhesive foam layer was used to hold each array of electrodes onto the skin, and conductive paste filled the cavities of the adhesive layers to make skin-electrode contact. For two-source validation of the EMG decomposition, an intramuscular linear array of 40 electrodes on a thin-film (platinum coated, interelectrode distance: 1mm) was inserted into the TA in one participant at an approximate angle of 30°. The insertion was guided with a portable ultrasound probe (Butterfly IQ+, Butterfly Network, USA). Two bands damped with water were placed around the ankle as ground and reference electrodes. EMG signals were recorded in monopolar derivation with a sampling frequency of 2048 Hz for surface electrodes, 10240 Hz for intramuscular electrodes, amplified (x150), band-pass filtered (10–500 Hz for surface; 100-4400 Hz for intramuscular), and digitized using a 400 channels acquisition system with a 16-bit resolution (EMG-Quattrocento; OT Bioelettronica, Italy).

## HD-EMG decomposition

The monopolar EMG signals were band-pass filtered between 20 and 500 Hz with a second-order Butterworth filter, and channels with low signal-to-noise ratio or artifacts were discarded after visual inspection. The EMG signals were decomposed into individual motor unit pulse trains using convolutive blind-source separation (Negro et al., 2016b [\[link\]](#)). EMG signals were first extended by adding delayed versions of each channel, with an extension factor, i.e., number of delayed versions, adjusted to reach a total of 1000 extended channels. The extended signals were spatially whitened to make them uncorrelated and of equal power. Thereafter, a fixed-point algorithm was applied to identify the sources of the EMG signals, i.e., the motor unit pulse trains. In this algorithm, a contrast function  $g(x) = \log(\cosh(x))$  and its derivatives were applied to the extended and whitened EMG signals to iteratively optimise a separation vector that skew the distribution of the values of the motor unit pulse trains toward 0, and thus maximise the level of sparsity of the motor unit pulse train. The high level of sparsity matches the physiological properties of motor units, with a relatively small number of discharges, or 1, per second (< 50 discharge times/s during submaximal isometric contractions). The convergence was reached once the level of sparsity did not substantially vary (with a tolerance fixed at 10<sup>-4</sup>) when compared to the previous iteration. At this stage, the motor unit pulse train contained high peaks (i.e., the discharge times of the identified motor unit) and lower values due to the activities of neighbouring motor units and noise. High peaks were separated from lower values using the MATLAB functions *findpeaks.m* and *kmeans.m* (with two classes). The peaks from the class with the highest centroid were considered as the discharge times of the identified motor unit.

After this automatic identification of the discharge times, duplicates were automatically removed. For this purpose, the pulse trains of pairs of motor units were first aligned using a cross-correlation function to account for a potential delay due to the propagation time of action potentials along the fibres. Then, two discharge times identified from these pulse trains were considered as common when they occurred within a time interval of 0.5 ms, and two or more motor units were considered as duplicates when they had at least 30% of their identified discharge times in common (Holobar et al., 2010 [DOI](#); Negro et al., 2016b [DOI](#)). In principle, the limited level of synchronisation between individual motor units results in a few simultaneous discharge times between pairs of motor units. A threshold of 30% is therefore highly conservative and ensure the removal of all motor units with a level of synchronisation well above physiological values. When duplicates were identified, the motor unit with the lowest coefficient of variation of the inter-spike intervals was retained for the analyses.

At the end of these automatic steps, all the motor unit pulse trains and identified discharge times were visually inspected, and manual editing was performed to correct the false identification of artifacts or the missed discharge times (Del Vecchio et al., 2020 [DOI](#); Hug et al., 2021 [DOI](#); Avrillon et al., 2023 [DOI](#)). The manual editing consisted of i) removing the spikes causing erroneous discharge rates (outliers), ii) adding the discharge times clearly separated from the noise, iii) recalculating the separation vector, iv) reapplying the separation vector on the entire EMG signals, and v) repeating this procedure until the selection of all the discharge times is achieved. The manual editing of potential missed discharge times and falsely identified discharge times was never immediately accepted. Instead, the procedure was consistently followed by the application of the updated motor unit separation vector on the entire EMG signals to generate a new motor unit pulse train. Then, the manual editing was only accepted when the silhouette value increased or stayed well above the threshold of 0.9 quantified with the silhouette value (Negro et al., 2016b [DOI](#)). Only these motor units were retained for further analysis.

## Validation of the decomposition

EMG decomposition was validated using both simulation of EMG signals and two-source validation. For the simulation, we generated the series of discharge times of a group of 150 motor units in an anatomical model entailing a cylindrical muscle volume with parallel fibres [see (Konstantin et al., 2020) for a full description of the model], in which subcutaneous and skin layers separate the muscle from the surface electrodes. We set the radius of the muscle to 15 mm and the thicknesses of the subcutaneous and skin layers to 4 mm and 1 mm, respectively. The motor unit action potentials were detected in the model by a grid of 64 circular surface electrodes with a diameter of 1 mm arranged in 5 columns and 13 rows (inter-electrode distance: 4mm). Following this, we estimated the rate of agreements between simulated and estimated series of motor unit discharge times, calculated as the ratio between correctly identified discharge times and the sum of correctly identified discharge times, missed discharge times, and falsely identified discharge times.

For the two-source validation, EMG signals were simultaneously recorded with an intramuscular linear array of 40 electrodes on a thin film and two grids of 64 surface electrodes. After the decomposition of both intramuscular and surface EMG signals, identified motor units were matched following the same procedure as for the elimination of duplicates. In short, the motor unit pulse trains were first aligned using a cross-correlation function. Two motor units identified with intramuscular and surface EMG decompositions were considered as matches when they had at least 30% of their discharge times in common. We calculated the accuracy of the decomposition using the rate of agreements between their series of discharge times.

## Proportion of the EMG signal represented by the decomposition

A synthetic EMG signal was reconstructed from the cumulative firing activity of the identified motor units. First, EMG signals from the 256 electrodes were differentiated in the column direction to obtain 236 single-differential EMG channels. Second, each differentiated EMG signal was segmented over successive windows of 25 ms centred around the discharge times of a motor unit. All the windows were averaged to estimate the average action potential waveform of that motor unit over the 236 each channel. The action potentials were then convolved with the series of discharge times to obtain trains of action potentials. These steps were repeated for all the identified motor units, and all the trains of action potentials were summed to reconstruct the synthetic EMG signals. At the end of this process, the ratio between the powers of synthetic and original EMG signals was calculated.

## Motor unit tracking

As explained in the section that described the decomposition framework and in previous studies (Francic and Holobar, 2021 [↗](#); Skarabot et al., 2023 [↗](#); Škarabot et al., 2023 [↗](#)), a motor unit pulse train results from the projection of extended and whitened EMG signals onto a separation vector, optimised during a fixed-point algorithm. Discharge times are automatically identified from this motor unit pulse train, before visually inspection and manual editing.

In this study, motor units were tracked by slightly adapting this process. First, EMG signals from two successive contraction levels, say 10% and 20% MVC, were separately decomposed. The motor unit pulse trains, the identified discharge times, and the associated separation vectors were saved for each contraction level. Second, the EMG signals from the highest contraction level (20% MVC) were projected onto the separation vectors identified in the lowest contraction level (10% MVC). A new group of pulse trains was generated, which represented the firing activity of all the motor unit identified at 10% MVC, but during the contraction performed at 20% MVC. When the high peaks were clearly separated from the noise, the discharge times were automatically identified using the MATLAB functions *findpeaks.m* and *kmeans.m*. Third, these new series of discharge times were matched with those that have been initially identified at 20% MVC using the same approach as during two-source validation (>30% of common discharge times, see above). This process ended with three groups of motor units: i) the motor units only identified at 10% MVC, ii) the motor units identified at 10% and 20% MVC, and iii) the motor units only identified at 20% MVC. These steps were repeated between all the successive contraction levels, that is, from between 20% and 30% MVC to between 70% and 80% MVC. At the end of this iterative process, the following data were saved for each motor unit across the contraction levels where they were tracked: the instantaneous firing rates and force during ramp up and ramp down phases (Figure 2 [↗](#)), the average firing rate during the plateau (Figure 3 [↗](#)), the recruitment threshold (Figure S3 [↗](#)).

## Motor unit uniqueness

The accuracy of motor unit tracking was further tested by confirming the uniqueness of each motor unit within the entire sample, assuming that each motor unit must have a unique representation of their action potentials across the array of surface electrodes (Figure 1 [↗](#); (Farina et al., 2008 [↗](#))). This was accomplished by calculating the root mean-square error (RMSE) between their action potentials across contractions and the action potentials of the rest of the motor units. EMG signals from the 256 electrodes were differentiated in the column direction to obtain 236 single-differential EMG channels. For each motor unit, the single-differential EMG signals were segmented around the discharge times and averaged to identify an average action potential waveforms for each channel. The action potential waveforms were concatenated in a matrix of 236 rows (EMG channels) and 102 columns (time samples; 50 ms). This process was repeated for each contraction level where the motor unit was tracked. A first RMSE was calculated between the concatenated action potentials identified across contractions, and this value was defined as a

reference. Then, RMSE was calculated between the concatenated action potentials of that motor unit and the concatenated action potentials of the rest of the motor units. A reference value lower than the 5<sup>th</sup> percentile of the distribution of RMSE calculated between motor units demonstrated that its distribution of action potentials across contractions were more similar than with the rest of the motor units, proving their uniqueness within the entire sample.

## Input-output function of motoneurons

The input-output function of each motor unit was characterised as the relation between its instantaneous firing rate and the muscle force. The instantaneous firing rates and muscle forces recorded during all the ramp-up phases where that motor unit was tracked were pooled. For example, the motor unit displayed in **Figure 2A** was tracked from 20% to 60% of the MVC. The relation between firing rates and force was characterised by comparing three curve fits:

- A linear fit:  $\text{Firing rate}(\text{Force}) = a(\text{Force}) + b$
- A rising exponential:  $\text{Firing rate}(\text{Force}) = a(1 - e^{-(\text{Force}/b)}) + c$
- A natural logarithm:  $\text{Firing rate}(\text{Force}) = a * \ln(\text{Force}) + b$

The parameters of the functions were estimated with the MATLAB functions *fitttype.m* and *fit.m* using a nonlinear least-squares solver with a maximum of 1000 iterations. The best model of the firing rate-force relation during the ramp up phase was the fit with the lowest Bayesian information criterion (BIC). This criterion assesses the performance of a model by balancing its goodness of fit with its complexity (number of parameters). The BIC was calculated as follow:

$$\text{BIC} = n * \ln(\text{sse} / n) + p * \ln(n)$$

where n is the number of data samples on which the fit was estimated, sse is the sum of squares error, and p is the number of parameters used in the model.

The input-output functions of motor units during the ramp-down phase were characterised using the same approach, and we reported in the results whether the best fit changed between the ramp-up and ramp-down phases.

## Rate coding of motor units

Most of motor units input-output functions were better characterised with a natural logarithm function. This function determines two stages in rate coding: a steep acceleration of firing rate followed by a slower linear increase in firing rate. Therefore, these two stages of the firing rate-force relation were further analysed. To estimate the acceleration of firing rate during the ramp-up phase, the first derivative of the firing rate-force relation fitted with a natural logarithm was computed as:

$$\text{Acceleration of firing rate}(\text{Force}) = a / \text{Force}$$

where a is the parameter of the equation of the natural logarithm function. The initial acceleration of firing rate was calculated as the value of the derivative at the recruitment threshold of the motor unit. The relation between the initial acceleration of firing rate and the recruitment threshold of all motor units across the pool was characterised for each muscle and each participant using nonlinear least-squares solvers in MATLAB. The slow linear increase in firing rate that followed the initial steep acceleration of firing rate was characterised by comparing the average firing rate of motor units during the plateaus of force between successive contraction levels (**Figure 3**). Specifically, the rate of increase in firing rate for an increment of 10 %MVC were calculated for each motor unit, and the relation between the rate of increase and the

recruitment threshold of all motor units across the pool was characterised for each participant and each muscle using a linear fit in MATLAB. Pearson's  $r$  and  $p$  values of these relations were reported in the results.

## Hysteresis between recruitment and derecruitment thresholds

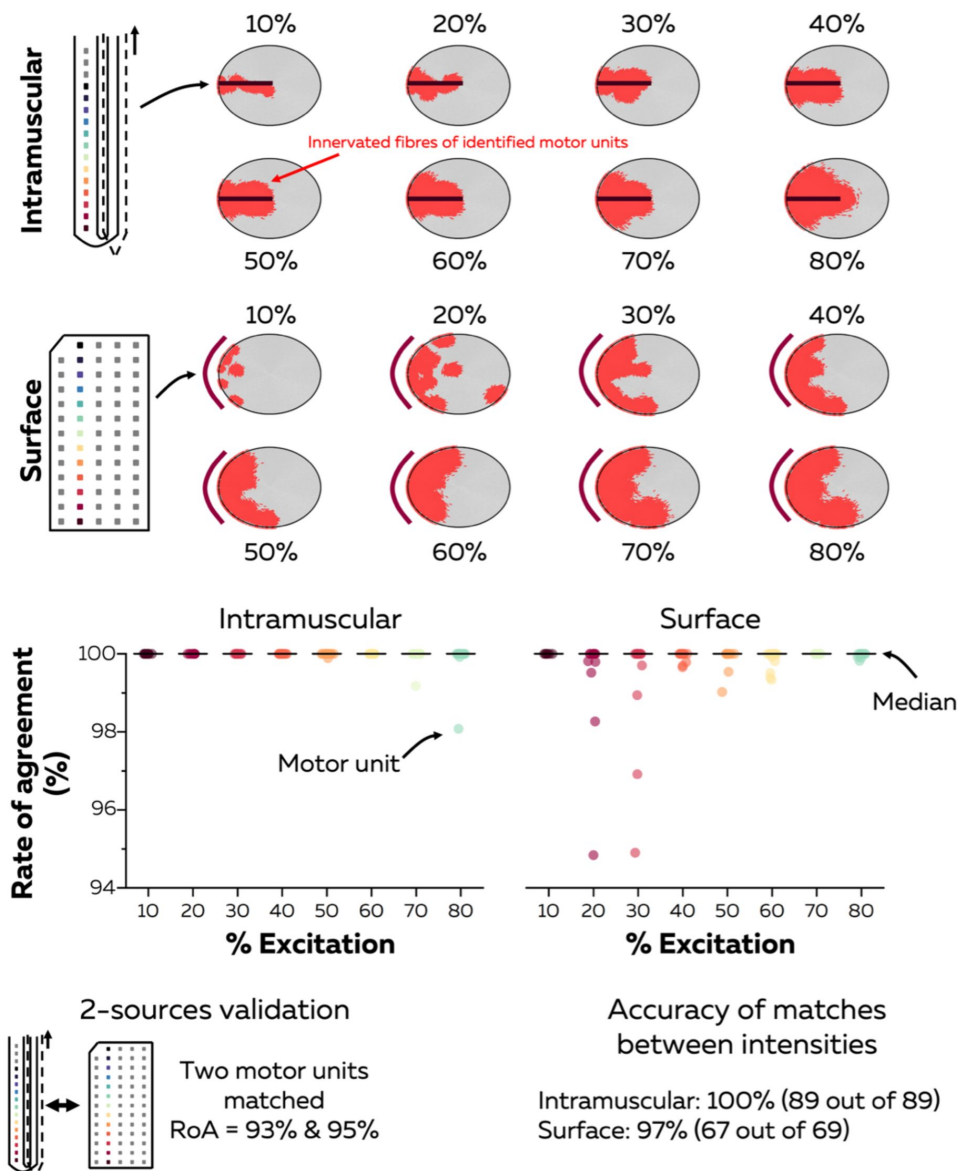
The relation between recruitment and derecruitment thresholds of all motor units across the pool were characterised for each participant and each muscle using non-linear and linear fits. As for the relations between the firing rate and force of each motor unit, the best fit was selected using the Bayesian information criterion. The differences between the recruitment and derecruitment thresholds (hysteresis) were calculated for each motor unit, with negative values indicating a hysteresis and a positive value indicating a reverse hysteresis.

## Statistical analyses

All statistical analyses were performed with RStudio (USA). Quantile-quantile plots and histograms were displayed to check the normality of the data distribution. If the distribution was determined not to be normal, they were transformed to remove the skew. All statistical analyses were performed using linear mixed effect models implemented in the R package *lmerTest* with the Kenward-Roger method to estimate the denominator degrees of freedom and the  $p$ -values. This method considers the dependence of data points (i.e., individual motor unit) in each participant. When necessary, multiple comparisons were performed using the R package *emmeans*, which adjusts the  $p$ -value using the Tukey or the Bonferroni method. The significance level was set at 0.05. Values are reported as mean  $\pm$  standard deviation.

Initial accelerations of firing rate and rates of increase in firing rate between successive contraction levels were compared between motor units using a linear mixed effect model, with *Muscle* (TA; VL) and *Threshold* (low; medium; high) as fixed effects and *Participant* as a random effect.

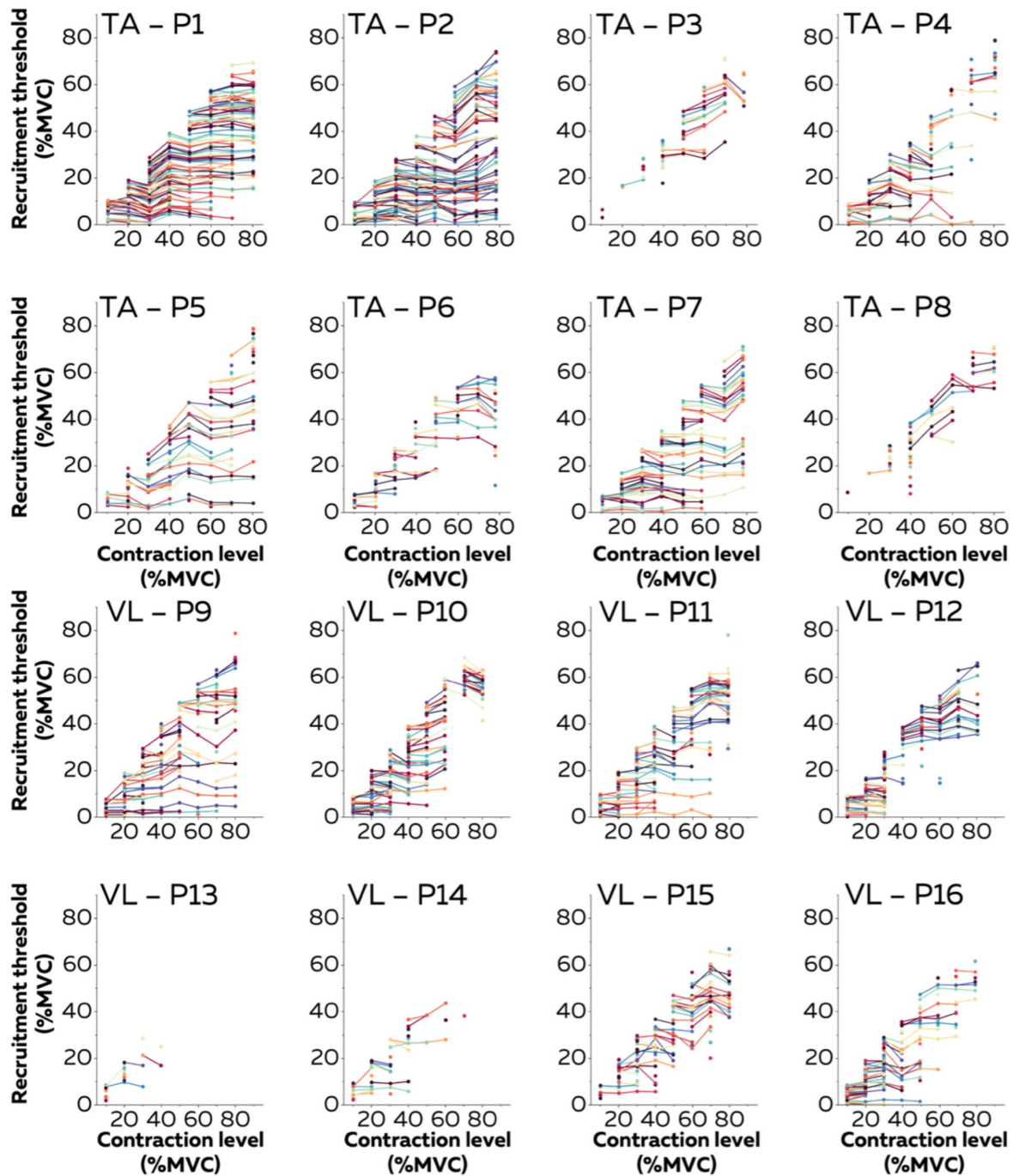
The difference between motor units' recruitment and derecruitment thresholds were compared using a linear mixed effect model, with *Muscle* (TA; VL) and *Threshold* (low; medium; high) as fixed effects and *Participant* as a random effect. Whether the difference between the two thresholds were significantly different from 0 (absence of hysteresis) was further tested using function *contrast* in the package *emmeans*.



**Figure S1.**

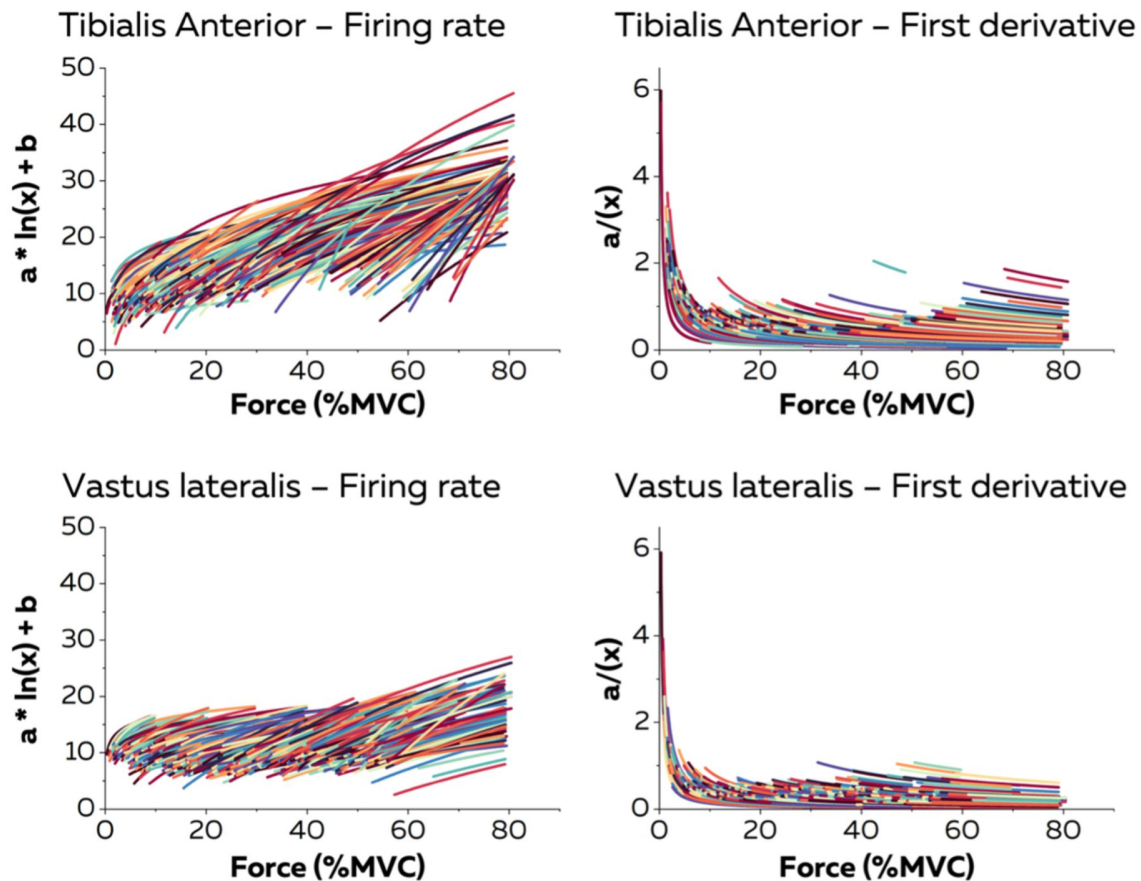
Validation of electromyographic decomposition and motor unit tracking. We used two approaches to validate the electromyographic decomposition. First, we simulated a pool of 150 motor units producing forces from 10 to 80% of the maximal excitation. Electromyographic signals were simulated from either 16 intramuscular electrodes (black horizontal line, top panels) or 64 surface electrodes (dark red curves on the left of top panels). Each of the grey elliptical shapes represents the cross-sectional area of the simulated muscle. We identified motor units using blind-source separation, with their territories displayed in red in the top panels. We then calculated the rate of agreement between the series of discharge times identified from the decomposition and simulated. Each data point is a motor unit, and the horizontal black line depicts the median. After this, we used the two-source validation approach, i.e., the comparison of discharge times identified from intramuscular and surface electromyographic signals. A participant performed trapezoidal isometric contractions with plateaus ranging from 10 to 80% of the maximal force while we recorded surface signals with a grid of 128 electrodes and intramuscular signals with 40 electrodes along a thin film that was inserted into the muscle with a hypodermic needle. Intramuscular and surface electromyographic signals were both decomposed and two motor units were matched between intramuscular and surface signals. We calculated the rate of agreement between the series of discharge times. It is worth noting that we identified the same number of discharge times with both methods. A rate of agreements less than 100% reflects misalignments of some discharge times with the surface signals due to the propagation of the action potentials through the tissue and potential overlapping action potentials at the surface of the skin. Finally, we validated the tracking method using the simulated signals, with high percentages highlighting the high accuracy of our approach.





**Figure S2.**

Recruitment thresholds of motor units tracked across contractions. We tracked motor units across contractions using their unique distribution of action potentials within the grid of 256 electrodes. Then, we estimated their recruitment thresholds as the average of force over a window of 10 ms centred around the first discharge time. The recruitment thresholds, estimated for each contraction where the motor unit was identified, are displayed for each participant. A series of data points connected with a line of the same colour is a motor unit. TA. Tibialis Anterior. VL. Vastus Lateralis. MVC. Maximal voluntary contraction level P1. Participant #1.



**Figure S3.**

Relations between force and firing rates fitted with a natural logarithm. We reported the relations between instantaneous motor units firing rates and the applied force. Then, we fitted these relations for each motor unit using a natural logarithm function (left panels). The non-linearities in the relations depict the influence of neuromodulatory inputs to motoneurons on the transformation between the synaptic inputs and the motor unit firing rates. To quantify the impact of the amplification effect of these neuromodulatory inputs, we estimated the acceleration of firing rate as the first derivative of the fitted firing rate/force curves (right panels). The fits are drawn over the range of forces where the motor unit was firing during the ramp-up phases of the contractions. Each line is a motor unit. MVC. Maximal voluntary contraction level.

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## Editors

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**Reviewer #1 (Public review):**

Summary:

The Avrillon et al. explore the neural control of muscle by decomposing the firing activity of constituent motor units from the grid of surface electromyography (EMG) in the Tibialis (TA) Anterior and Vastus Lateralis (VL) during isometric contractions. The study involves extensive samples of motor units across the broadest range of voluntary contraction intensities up to 80% of MVC. The authors examine rate coding of the population of motor units, which describes the instantaneous firing rate of each motor unit as a function of muscle force. This relationship is characterized by a natural logarithm function that delineates two distinct phases: an initial phase with a steep acceleration in firing rate, particularly pronounced in low-threshold motor units, and a subsequent modest linear increase in firing rate, more significant in high-threshold motor units.

Strengths:

The study makes a significant contribution to the field of neuromuscular physiology by providing a detailed analysis of motor unit behavior during muscle contractions in a few ways.

(1) The significance lies in its comprehensive framework of motor unit activity during isometric contractions in the broad range of intensities, providing insights into the non-linear relationship between the firing rate and the muscle force. The extensive sample of motor units across the pool confirms the observation in animal studies in which the the spinal motoneuron exhibits a discharge consists of the distinct phases in response to synaptic currents, under the influence of persistent inward currents. As such, it is now reasonable to state the human motor units across the pool are also under control of gain modulation via some neuromodulatory effects in addition to synaptic inputs arising from ionotropic effects. (2) The firing scheme across in the entire motoneuron pool revealed in this study reconciles the discrepancy in firing organization under debate; i.e., whether it is 'onion skin' like or not (Heckman and Enoka 2012). The onion skin like model states that the low threshold motor units discharge higher than high threshold motor units and has been held for long time because the firing behaviors were examined in a partial range of contraction force range due to technical limitations. This reconciliation is crucial because it is fundamental to modelling the organization of motor unit recruitment and rate coding to achieve a desired force generation to advance our understanding of motor control.

(3) The extensive data collection with a novel blind source separation algorithm on the expanded number of channel of surface EMG signal provides a robust dataset that enhances the reliability and validity of findings, setting a new standard for empirical studies in the field.

Collectively, this study fills several knowledge gaps in the field and advances our understanding the mechanism underlying the isometric force generation.

<https://doi.org/10.7554/eLife.97085.2.sa3>

**Reviewer #2 (Public review):**

Avrillon et al. provides a comprehensive assessment of firing rate parameters from a large percentage of the motor unit pool, in two muscles, during voluntary isometric contractions. The authors have used new quantitative methods to extract more unique motor units across

contractions than prior studies. This was achieved by recording muscle fibre action potentials from four high density surface electromyogram (HDsEMG) arrays, quantifying residual EMG comparing the recorded and data-based simulation (Fig. 1A-B), and developing a metric to compare the spatial identification for each motor unit (Fig. 1D-E). From identified motor units, the authors have provided a detailed characterization of recruitment and firing rate responses during slow voluntary isometric contractions in the vastus lateralis and tibialis anterior muscles up to 75-80% of maximum intensity. In the lower limb it is interesting how lower threshold motor units have firing rate responses that saturate, whereas higher threshold units that presumably produce higher muscle contractile forces continue to increase their firing rate. Conceptually, the authors rightly focus on the literature of intrinsic motoneurone properties, but in vivo, other possibilities (that are difficult to measure in awake human participants) are that the form of descending supraspinal drive, spinal network dynamics and afferent inputs may have different effects across motor unit sizes, muscles and types of contractions. These results from single trial contractions and with a larger sample of motor units, supports the summary rate coding profiles of motor units in the extensor digitorum communis muscle (Monster and Chan, 1977).

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### Reviewer #3 (Public review):

#### Summary:

This is an interesting manuscript which uses state of the art experimental and simulation approaches to quantify motor unit discharge patterns in the human TA and VL. The non-linear profiles of motor unit discharge were calculated and found to have an initial acceleration phase followed by an attenuation phase. Lower threshold motor units had a larger gain of the initial acceleration whereas the higher threshold motor unit had a higher gain in the attenuation phase. These data represent a technical feat and are important for understanding how humans generate and control voluntary force.

#### Strengths:

The authors used rigorous, state-of-the art analyses to decompose and validate their motor unit data during a wide range of voluntary efforts.

Analyses are clearly presented, applied, and visualized.

The supplemental data provides important transparency.

#### Weaknesses:

Number of participants and muscles tested are relatively small - particularly given the constraints on yield. It is unclear if this will translate to other motor pools. The justification for TA and VL should be provided.

While in impressive effort was made to identify and track motor units across a range of contractions, it appears that a substantial portion of muscle force was not identified. Though high intensity contractions are challenging to decompose - the authors are commended in their technical ability in recording population motor unit discharge times with recruitment thresholds up to 75% a participant's maximal voluntary contractions. However previous groups have seen substantial recruitment motor units above 80% and even 90% maximum activation in the soleus. Given the innervation ratios of higher threshold motor units, if recruitment continued to 100%, the top quartile would likely represent a substantial portion of the traditional fast-fatigable motor units. It would be highly interesting to understand the

recruitment and rate coding of the highest threshold motor units, at a minimum I would suggest using terms other than "entire range" or "full spectrum of recruitment thresholds"

The quantification of hysteresis using torque appears to make self-evident the observation that lower threshold motor units demonstrate less hysteresis with respect to torque - If there was motor unit discharge there will be force. I believe this limitation goes beyond the floor effects discussed in the manuscript. Traditionally individuals have used the discharge of a lower threshold unit as the measure on which to apply hysteresis analyses to infer ion channel function in human spinal motoneurons.

The main findings are not entirely novel. See Monster and Chan 1977 and Kanosue et al 1979

Comments on revisions:

I thank the authors for their thoughtful revision.

Just to confirm, the ranges for motor unit yield are for a single contraction. So, for example, in a participant there were 71 unique and concurrently active VL motor units able to be decomposed.

<https://doi.org/10.7554/eLife.97085.2.sa1>

#### Author response:

The following is the authors' response to the original reviews.

##### **Reviewer #1 (Public Review):**

###### *Summary:*

*This study explores the neural control of muscle by decomposing the firing activity of constituent motor units from the grid of surface electromyography (EMG) in the Tibialis (TA) Anterior and Vastus Lateralis (VL) during isometric contractions. The study involves extensive samples of motor units across the broadest range of voluntary contraction intensities up to 80% of MVC. The authors examine the rate coding of the population of motor units, which describes the instantaneous firing rate of each motor unit as a function of muscle force. This relationship is characterized by a natural logarithm function that delineates two distinct phases: an initial phase with a steep acceleration in firing rate, particularly pronounced in low-threshold motor units, and a subsequent modest linear increase in firing rate, more significant in high-threshold motor units.*

###### *Strengths:*

*The study makes a significant contribution to the field of neuromuscular physiology by providing a detailed analysis of motor unit behavior during muscle contractions in a few ways.*

*(1) The significance lies in its comprehensive framework of motor unit activity during isometric contractions in a broad range of intensities, providing insights into the non-linear relationship between the firing rate and the muscle force. The extensive sample of motor units across the pool confirms the observation in animal studies in which the spinal motoneuron exhibits a discharge consisting of distinct phases in response to synaptic currents, under the influence of persistent inward currents. As such, it is now reasonable to state the human motor units across the pool are also under the control of gain modulation via some neuromodulatory effects in addition to synaptic inputs arising from ionotropic effects.*

(2) The firing scheme across the entire motoneuron pool revealed in this study reconciles the discrepancy in firing organization under debate; i.e., whether it is 'onion skin' like or not (Heckman and Enoka 2012). The onion skin like model states that the low threshold motor units discharge higher than high threshold motor units and have been held for a long time because the firing behaviors were examined in a partial range of contraction force range due to technical limitations. This reconciliation is crucial because it is fundamental to modelling the organization of motor unit recruitment and rate coding to achieve a desired force generation to advance our understanding of motor control.

(3) The extensive data collection with a novel blind source separation algorithm on the expanded number of channels of surface EMG signal provides a robust dataset that enhances the reliability and validity of findings, setting a new standard for empirical studies in the field.

Collectively, this study fills several knowledge gaps in the field and advances our understanding of the mechanism underlying the isometric force generation.

We thank the reviewer for their positive appreciation of our work.

#### Weaknesses:

Although the findings and claims based on them are mostly well aligned, some accounts of the methods and claims need to be clarified.

(1) The authors examine the input-output function of a motor unit by constructing models, using force as an input and discharge rate as an output. It sounds circular, or the other way around to use the muscle force as an input variable, because the muscle force is the result of motor unit discharges, not the cause that elicits the discharges. More specifically, as a result of non-linear interactions of synchronous and/or asynchronous discharges of a population of a given motoneuron pool that give rise to transient increase/maintenance in twitch force, the gross muscle force is attained. I acknowledge that it is extremely challenging experimentally to measure synaptic currents impinging upon the spinal motoneurons in human subjects and the author has an assumption that the force could be used as a proxy of synaptic currents. However, it is necessary to explicitly provide the caveats and rationale behind that. Force could be used as the input variable for modelling.

Force is indeed used in this study as a proxy of the common excitatory synaptic currents as their direct measurement is not possible in vivo in humans. It is worth noting that this approach has been extensively used in the past by many groups to study rate coding (e.g., Monsters & Chan, De Luca's, Heckman's, and Fuglevand's groups). Heckman's, Gorassini's, Fuglevand's groups and others have considered the non-linearities in the relation between motor unit firing rates and muscle force in humans as an indicator of the impact of neuromodulation on motor unit behaviour and changes of the intrinsic properties of motoneurons.

One could also use the cumulative spike train as a more direct estimate of common excitatory inputs, assuming that it is possible to identify a group of motor units not influenced by PICs, as done when selecting a reference low-threshold motor neuron in the delta F method (Gorassini et al., 1998), or the cumulative spike train of low-threshold motor neurons (Afsharipour et al., 2020). However, this approach was not possible in our study as we did not have the same units across contractions to estimate cumulative spike trains. It was therefore not possible to pool the data across contractions as we did to generate force/firing rate relations on the widest range of force.



We added a sentence in the discussion to highlight this limitation (P19, L470):

‘This result must be confirmed with a more direct proxy of the net synaptic drive, such as the firing rate of a reference low-threshold motor neuron used in the delta F method (Gorassini et al., 1998), or the cumulative spike train of low-threshold motor neurons (Afsharipour et al., 2020)’.

*(2) The authors examine the firing organizations in TA and VL in this study without explicit purposes and rationale for choosing these muscles. The lack of accounts makes it hard for the readers to interpret the data presented, particularly in terms of comparing the results from the different muscles.*

We wanted to compare the rate coding of pools of motor units from proximal (VL) and distal (TA) muscles within the lower limb. Indeed, distal and proximal muscles exhibit differences in rate coding and spatial recruitments (De Luca et al., 1982, J Physiol), potentially due to different levels of recurrent inhibition (Cullheim & Kellerth, 1978, J Physiol; Rossi & Mazzocchio, 1991, Exp Brain Res; Edgley et al., 2021, J Neurosci) or different levels of neuromodulation depending on their involvement (or not) in postural control (Hoonsgaard et al., 1988, J Physiol; Kim et al., 2020, J Neurophysiol).

We added a paragraph at the beginning of the result section to support our muscle choice (P6; L137): ‘16 participants performed either isometric dorsiflexion (n = 8) or knee extension tasks (n = 8) while we recorded the EMG activity of the tibialis anterior (TA - dorsiflexion) or the vastus lateralis (VL – knee extension) with four arrays of 64 surface electrodes (256 electrodes per muscle). The motoneuron pools of these two muscles of the lower limb receive a large part of common input (Laine et al., 2015; Negro et al., 2016a), constraining the recruitment of their motor units in a fixed order across tasks. They are therefore good candidates for an accurate description of rate coding. Moreover, we wanted to determine whether differences in rate coding observed between proximal and distal muscles in the upper limb (De Luca et al., 1982) were also present in the lower limb.’

Another factor that guided our muscle choice was the low risk of crosstalk. For this, we verified with ultrasound that our arrays of 256 electrodes only covered the muscle of interest, staying away from the neighbouring muscles. This was possible as superficial muscles from the leg are bulkier than those from the upper limb. Given the small diameter of each electrode (2 mm), it is unlikely that the motor units from the neighbouring muscles were in the recorded muscle volume (Farina et al., 2003, IEEE Trans Biomed Eng)

*(3) In the methods, the author described the manual curation process after applying the blind source separation algorithm. For the readers to understand the whole process of decomposition and to secure rigor and robustness of the analyses, it would be necessary to provide details on what exact curation is performed with what criteria.*

The manual curation of EMG decomposition with blind source separation is different from what is classically done with intramuscular EMG and template-matching algorithms.

In short, our decomposition algorithm uses fast independent component analysis (fastICA) to retrieve motor unit spike trains from the EMG signals. For this, it iteratively optimises a set of weights, i.e., a separation vector, for each motor unit. The projection of the EMG signals on this separation vector generates a sparse motor unit pulse train, with most of its samples close to zero and only a few samples close to one (Figure 1B). The discharge times are estimated from this motor unit pulse train using a peak detection function and a k-mean classification with two classes to separate the high peaks (spikes) from the low peaks (noise and other motor units).

The manual curation consists of inspecting the automatic detection of the peaks of the motor unit pulse train and manually add missed peaks (missed discharge times) or remove wrongly detected peaks. Then, the separation vector is updated using the correct discharge times and the motor unit pulse train recalculated. This procedure generally improves the distance between the discharge times and the noise, which confirm the accuracy of the manual curation. If that's not the case, the motor unit is discarded from the analyses.

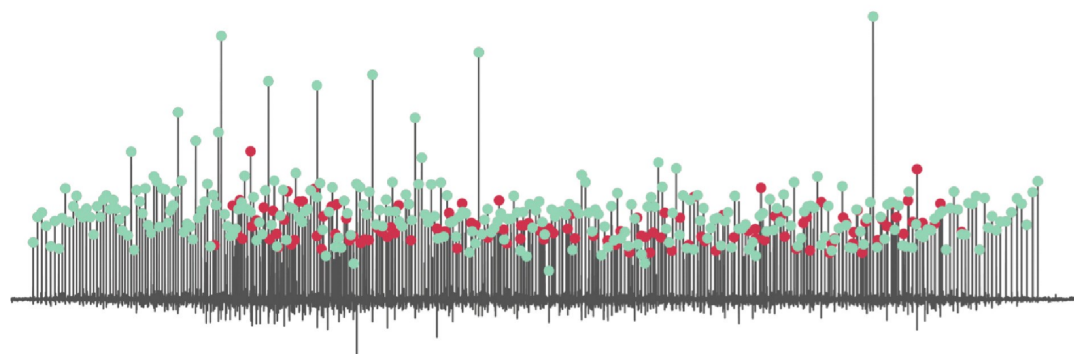
We added a section on manual editing in the methods (P23, L615):

‘At the end of these automatic steps, all the motor unit pulse trains and identified discharge times were visually inspected, and manual editing was performed to correct the false identification of artifacts or the missed discharge times (Del Vecchio et al., 2020; Hug et al., 2021; Avrillon et al., 2023). The manual editing consisted of i) removing the spikes causing erroneous discharge rates (outliers), ii) adding the discharge times clearly separated from the noise, iii) recalculating the separation vector, iv) reapplying the separation vector on the entire EMG signals, and v) repeating this procedure until the selection of all the discharge times is achieved. The manual editing of potential missed discharge times and falsely identified discharge times was never immediately accepted. Instead, the procedure was consistently followed by the application of the updated motor unit separation vector on the entire EMG signals to generate a new motor unit pulse train. Then, the manual editing was only accepted when the silhouette value increased or stayed well above the threshold of 0.9 quantified with the silhouette value (Negro et al., 2016b). Only these motor units were retained for further analysis.’

*(4) In Figure 3, the early recruited units tend to become untraceable in the higher range of contraction. This is more pronounced in the muscle VL. This limitation would ambiguate the whole firing curve along the force axis and therefore limitation and the applicability in the different muscles needs to be discussed.*

The loss of low threshold motor units in the higher range of contractions was caused either by the decrease in signal-to-noise ratio for small motor units when many larger ones are recruited, or by the cancellation of the surface action potentials of the small units in the interference electromyographic signal, or by the recruitment of a motor unit with a very similar spatio-temporal filter (an example is shown in the figure below). In the latter case, the motor unit pulse train contains peaks that represent the discharge times of both motor units (green and red dots in the simulated example below), making them undistinguishable by the operator during manual editing.

**Author response image 1.**



This was discussed in the results (P7; L190):

‘On average, we tracked  $67.1 \pm 10.0\%$  (25th–75th percentile: 53.9 – 80.1%) of the motor units between consecutive contraction levels (10% increments, e.g., between 10% and 20% MVC) for TA and  $57.2 \pm 5.1\%$  (25th–75th percentile: 46.6 – 68.3%) of the motor units for VL (Figure S2). There are two explanations for the inability to track all motor units across consecutive contraction levels. First, some motor units are recruited at higher targets only. Second, it is challenging to track small motor units beyond a few contraction levels due to a lower signal-to-noise ratio for the small motor units when larger motor units are recruited, or signal cancellation (Keenan et al., 2005; Farina et al., 2014a).’

However, we believe that it had a limited impact on the output of the paper, as the non-linear portion of the rate coding/force relation due to the persistent inward currents occurs during the first seconds after recruitment, before plateauing (for a review see Binder et al., 2020, Physiology).

*(5) It is unclear how commonly the notion "the long-held belief that rate coding is similar across motor units from the same pool" is held among the community without a reference. Different firing organizations have been modelled and discussed in the seminal paper by Fuglevand et al. (1993) and as far as I understand, the debate has not converged to a specific consensus. As such, any reference would be required to support the claim the notion is widely recognized.*

In the paper of Fuglevand et al., (1993, J Neurophysiol), all the motor units had the same rate coding pattern relative to the excitatory input, though they changed the slope of the relations and the saturation threshold of motor units between simulations. This is similar to the paper of De Luca & Contessa (2012, J Neurophysiol), where the equation used to simulate the rate coding was non-linear, but consistent across motor units.

We added these citations to the text:

‘Overall, we found that motor units within a pool exhibit distinct rate coding with changes in force level (Figure 2 and 3), which contrasts with the long-held belief that rate coding is similar across motor units from the same pool (Fuglevand et al., 1993; De Luca and Contessa, 2012).’

*(6) The authors claim that the firing behavior as a function of force is well characterized by a natural logarithmic function, which consists of initial steep acceleration followed by a modest increase in firing rate. Arguably the gain modulation in firing rate could be attributed to a neuromodulatory effect on the spinal motoneuron, which has been suggested by a number of animal studies. However, the complexity of the interactions between ionotropic and neuromodulatory inputs to motoneurons may require further elucidation to fully understand the mechanisms of neural control; it is possible to consider the differential acceleration among different threshold motor units as a differential combinatory effect of ionotropic and neuromodulatory inputs, but it is not trivially determined how differentially or systematically the inputs are organized. Likewise, the authors make an account for the difference in firing rate between TA and VL in terms of different amounts or balances of excitatory and inhibitory inputs to the motoneuron pool, but again this could be explained by other factors, such as a different extent of neuromodulatory effects. To determine the complexity of the interactions, further studies will be warranted.*

We appreciate the reviewer’s view on this point, as we indeed only indirectly inferred the combination of neuromodulatory and ionotropic inputs to motoneurons in this study. A more direct manipulation of the sources of neuromodulatory and ionotropic inputs will be required in the future to directly highlight the mechanisms responsible for these variations

in rate coding within pools. However, it is also worth noting that the acceleration in firing rate, the increase in firing rate during the ramp up, and the hysteresis between ramps up and downs have been used to infer the distribution of ionotropic and neuromodulatory inputs from the firing rate/force relations (Johnson et al., 2017; Beauchamp et al., 2023; Chardon et al., 2023). This approach has been validated with hundreds of thousands of simulations using a biophysical model of motor neurons (Chardon et al., 2023). There is also a series of studies in humans showing how the absence of neuromodulation modulated via inhibitory inputs (Revill & Fuglevand, 2017) or medication blocking serotonin receptors (Goodlich et al., 2023) impact the non-linearity of the firing rate/force relation. Therefore, we are confident that the differences observed within and between pools are linked to different distribution of excitatory/inhibitory inputs and neuromodulation.

We added a sentence in the discussion to highlight this point (P18; L435):

‘Taken together, these results show how ionotropic and neuromodulatory inputs to motoneurons uniquely combine to generate distinct rate coding across the pool, even if a more direct manipulation of the sources of neuromodulatory and ionotropic inputs will be required to directly estimate their interactions.’

*(7) It is unclear with the account " ... the bandwidth of muscle force is < 10Hz during isometric contraction" in the manuscript alone, and therefore, it is difficult to understand the following claim. It appears very interesting and crucial for motor unit discharge and force generation and maintenance because it would pose a question of why the discharge rate of most motor units is higher than 10Hz, despite the bandwidth being so limited, but needs to be elaborated.*

We described the slow fluctuations in smoothed firing rates associated with the variations in force observed during isometric contractions. The bandwidth of muscle force is lower than 10Hz due to the contractile properties of muscle tissues (Baldissera et al., 1998, J Physiol). Having an average firing rate higher than this bandwidth enables the pool of motor neurons to effectively transmit the common inputs (the main discriminant of muscle force) over this bandwidth without distortion (Farina et al., 2014, J Physiol). Increasing the firing rate beyond the muscle bandwidth also increases the power of the spike train at the direct current frequency (frequency equal to 0) since this power is related to the number of spikes per second. Thus, increasing the firing rate well beyond the muscle bandwidth still has a clear effect in force. To illustrate this point, note that electrical stimuli delivered at 100 Hz can lead to an increase in muscle force.

#### **Reviewer #2 (Public Review):**

##### *Summary:*

*The motivation for this study is to provide a comprehensive assessment of motor unit firing rate responses of entire pools during isometric contractions. The authors have used new quantitative methods to extract more unique motor units across contractions than prior studies. This was achieved by recording muscle fibre action potentials from four high-density surface electromyogram (HDsEMG) arrays (Caillet et al., 2023), quantifying residual EMG comparing the recorded and data-based simulation (Figure 1A-B), and developing a metric to compare the spatial identification for each motor unit (Figure 1D-E). From identified motor units, the authors have provided a detailed characterization of recruitment and firing rate responses during slow voluntary isometric contractions in the vastus lateralis and tibialis anterior muscles up to 80% of maximum intensity. In the lower limb, it is interesting how lower threshold motor units have firing rate responses that saturate, whereas higher threshold units that presumably produce higher muscle contractile forces continue to increase their firing rate. In many ways,*

*these results agree with the rate coding of motor units in the extensor digitorum communis muscle (Monster and Chan, 1977). The paper is detailed, and the analyses are well explained. However, there are several points that I think should be addressed to strengthen the paper.*

We thank the reviewer for their positive appreciation of our work.

*General comments:*

*(1) The authors claim they have measured the complete rate coding profiles of motor units in the vastus lateralis and tibialis anterior muscles. However, this study quantified rate coding during slow and prolonged voluntary isometric contractions whereas the function of rate coding during movements (Grimby and Hannerz, 1977) or more complex isometric contractions (Cutsem and Duchateau, 2005; Marshall et al., 2022) remains unexplored. For example, supraspinal inputs may not scale the same way across low and higher threshold motor units, or between muscles (Devanne et al., 1997), making the response of firing rates to increasing isometric contraction force less clear.*

*We agree with the reviewer that rate coding strategies may vary with the velocity and the type of contractions (Duchateau & Enoka, 2008, J Physiol). It is thus likely that the firing rate would increase during the first milliseconds of fast contractions, with the occurrence of doublets (Cutsem and Duchateau, 2005, J Physiol; Del Vecchio et al., 2019, J Physiol), or that motor unit firing rate may be lower during lengthening than shortening contractions (Duchateau & Enoka, J Physiol).*

*However, the decomposition of EMG signals in non-stationary conditions remains challenging, and is still limited to slow varying patterns of force (Chen et al., 2000, Oliveira & Negro, 2021, Mendez Guerra et al., 2024, Yeung et al., 2024). Future methodological developments will be required to expand our findings to other patterns of force.*

*Conceptually, the authors focus on the literature on intrinsic motoneurone properties, but in vivo, other possibilities are that descending supraspinal drive, spinal network dynamics, and afferent inputs have different effects across motor unit sizes, muscles, and types of contractions. Also, the influence from local muscles that act as synergists (e.g., vastii muscles for the vastus lateralis, and peroneal muscles that evert the foot for the tibialis anterior) or antagonists (coactivation during higher contraction intensities would stiffen the joint) may provide differential forms of proprioceptive feedback across motor pools.*

The reviewer is right that differences in spinal network dynamics and afferent inputs may explain the differences in rate coding observed between the two muscles. Indeed, computational models have shown how the pattern of inhibitory inputs may affect the increase in firing rate during linear increase in force (Powers & Heckman, 2017, J Neurophysiol; Chardon et al., 2023, Elife). Specifically, the difference observed between proportional inhibitory inputs vs. a push pull pattern mirror the differences observed here between the TA (push-pull like pattern) and the VL (proportional pattern). This difference may reflect the impact of various pathways of inhibition, such as reciprocal inhibition or recurrent inhibition from homonymous motor units or motor units from synergistic muscles.

These points have been further discussed in the manuscript (P19; L475):

‘The increase in firing rate was also significantly greater for TA motor units than for those in VL. This difference may reflect a varying balance between excitatory/inhibitory synaptic

inputs and neuromodulation due to multiple spinal circuits (Heckman and Binder, 1993; Heckman et al., 2008; Johnson et al., 2017; Powers and Heckman, 2017; Chardon et al., 2023; Škarabot et al., 2023). Specifically, the strength of recurrent and reciprocal inhibitory inputs to motoneurons innervating VL and TA, and their proportional or inverse covariation with excitatory inputs, respectively, may explain the differences in rate limiting and maximal firing rates (Heckman and Binder, 1993; Heckman et al., 2008; Johnson et al., 2017; Powers and Heckman, 2017; Chardon et al., 2023; Škarabot et al., 2023). Thus, the motor units from the VL may receive more recurrent inhibition than those of distal muscles, though direct evidence of these differences remains to be found in humans (Windhorst, 1996). Interestingly, similar differences in rate coding were previously observed between proximal and distal muscles of the upper limb (De Luca et al., 1982). However, other muscles that serve different functions within the human body, such as muscles from the face, have different rate coding characteristics with much higher firing rates (Kirk et al., 2021). Future work should investigate those muscles and other to reveal the myriads of rate coding strategies in human muscles.'

*(2) The evidence that the entire motor unit pool was recorded per muscle is not clear. There appears to be substantial residual EMG (Figure 1B), signal cancellation of smaller motor units (lines 172-176), some participants had fewer than 20 identified motor units, and contractions never went above 80% of MVC. Also, to my understanding, there remains no gold-standard in awake humans to estimate the total motor unit number in order to determine if the entire pool was decomposed.*

The reviewer is right that we did not decode the full pool of motor units. As indicated in the initial version of the manuscript (e.g. title, introduction), we considered that we identified an extensive sample of motor units representative of the dynamic of the pool. This claim was supported by the identification of motor units with recruitment thresholds ranging from 0 to 75% of the maximal force.

This statement was in the introduction (P4; L109): 'We were able to identify up to ~200 unique active motor units per muscle and per participant in two human muscles in vivo, yielding extensive samples of motor units that are representative of the entire motoneuron pools (Cailliet et al., 2023a).'

*Furthermore, using four HDsEMG arrays also raises questions about how some channels were placed over non-target muscles, and if motor units were decomposed from surrounding synergists.*

A factor that guided our muscle choice was the low risk of crosstalk. For this, we verified with ultrasound that our arrays of 256 electrodes only covered the muscle of interest, staying away from the neighbouring muscles. This was possible as superficial muscles from the leg are bulkier than those from the upper limb. Given the small diameter of each electrode (2 mm), it is unlikely that the motor units from the neighbouring muscles were in the recorded muscle volume.

*(3) The authors claim (Abstract L51; Discussion L376) that a commonly held view in the field is that rate coding is similar across motor units from the same pool. Perhaps this is in reference to some studies that have carefully assessed lower threshold motor units during lower force ramp contractions (e.g., Fuglevand et al., 2015; Revill and Fuglevand, 2017). However, a more complete integration of the literature exploring motor unit firing rate responses during rapid isometric contractions, comparing different muscles and contraction intensities would be helpful. From Figure 3, the range of rate coding in the tibialis anterior (~7-40 Hz) is greater than the vastus lateralis (~5-22 Hz) muscle across contraction levels. In agreement with other studies, the range of rate coding within some*



*muscles is different than others (Kirk et al., 2021) and during maximal intensity (Bellemare et al., 1983) or rapid contractions (Desmedt and Godaux, 1978). Likewise, within a motor pool, there is a diversity of firing rate responses across motor units of different sizes as a function of isometric force (Monster and Chan, 1977; Desmedt and Godaux, 1977; Kukula and Clamann, 1981; Del Vecchio et al., 2019; Marshall et al., 2022). A strength of this paper is how firing rate responses are quantified across a wide range of motor unit recruitment thresholds and between two muscles. I suggest improving clarity for the general reader, especially in the motivation for testing two lower limb muscles, and elaborating on some of the functional implications.*

We thank the reviewer for his input on this question. We have added references to these works and lines of research in the discussion:

(P18; L449): ‘In addition, rate coding patterns should also vary with the pattern of contractions, with fast contractions lowering the range of recruitment thresholds within motoneuron pools (Desmedt and Godaux, 1977b, 1979; van Bolhuis et al., 1997). The variability in rate coding observed here between motor units from the same pool could lead to small deviations from the size principle sometimes observed between pairs of units during isometric contractions with various patterns of force (Desmedt and Godaux, 1979; Marshall et al., 2022) or during the derecruitment phase (Bracklein et al., 2022).’ (P19; L487): ‘However, other muscles that serve different functions within the human body, such as muscles from the face, have different rate coding characteristics with much higher firing rates (Kirk et al., 2021). Future work should investigate those muscles and other to reveal the myriads of rate coding strategies in human muscles.’

In addition to the responses above, we have added a section at the beginning of the results to motivate the choice of the muscles (P6; L137):

‘16 participants performed either isometric dorsiflexion ( $n = 8$ ) or knee extension tasks ( $n = 8$ ) while we recorded the EMG activity of the tibialis anterior (TA - dorsiflexion) or the vastus lateralis (VL - knee extension) with four arrays of 64 surface electrodes (256 electrodes per muscle). The motoneuron pools of these two muscles of the lower limb receive a large part of common input (Laine et al., 2015; Negro et al., 2016a), constraining the recruitment of their motor units in a fixed order across tasks. They are therefore good candidates for an accurate description of rate coding. Moreover, we wanted to determine whether differences in rate coding observed between proximal and distal muscles in the upper limb (De Luca et al., 1982) were also present in the lower limb.’

### **Reviewer #3 (Public Review):**

#### **Summary:**

*This is an interesting manuscript that uses state-of-the-art experimental and simulation approaches to quantify motor unit discharge patterns in the human TA and VL. The non-linear profiles of motor unit discharge were calculated and found to have an initial acceleration phase followed by an attenuation phase. Lower threshold motor units had a larger gain of the initial acceleration whereas the higher threshold motor unit had a higher gain in the attenuation phase. These data represent a technical feat and are important for understanding how humans generate and control voluntary force.*

#### **Strengths:**

*The authors used rigorous, state-of-the-art analyses to decompose and validate their motor unit data during a wide range of voluntary efforts.*

*The analyses are clearly presented, applied, and visualized.*

*The supplemental data provides important transparency.*

We thank the reviewer for their positive appreciation of our work.

*Weaknesses:*

*The number of participants and muscles tested are quite small - particularly given the constraints on yield. It is unclear if this will translate to other motor pools. The justification for TA and VL should be provided.*

One strength of our study is to provide relations between key-parameters of rate coding (acceleration in firing rate, increase in firing rate, hysteresis) and the recruitment thresholds of motor units within two different pools, and for each individual participant. These relations were consistent across all the participants (Figures 2 to 4), making us confident that increasing the sample size would not change the conclusions of the study.

It is likely that the differences observed here between the VL and TA will also appear between other muscles of the leg, due to differences in the arrays of excitatory and inhibitory inputs they receive, the pattern of inhibitory inputs during increases in force (recurrent/reciprocal inhibition), and different levels of neuromodulation (Johnson et al., 2017, J Neurophysiol; Beauchamp et al., 2023; J Neural Eng). We have added a paragraph in the results to motivate our choice of muscles (P6; L137):

‘16 participants performed either isometric dorsiflexion (n = 8) or knee extension tasks (n = 8) while we recorded the EMG activity of the tibialis anterior (TA - dorsiflexion) or the vastus lateralis (VL – knee extension) with four arrays of 64 surface electrodes (256 electrodes per muscle). The motoneuron pools of these two muscles of the lower limb receive a large part of common input (Laine et al., 2015; Negro et al., 2016a), constraining the recruitment of their motor units in a fixed order across tasks. They are therefore good candidates for an accurate description of rate coding. Moreover, we wanted to determine whether differences in rate coding observed between proximal and distal muscles in the upper limb (De Luca et al., 1982) were also present in the lower limb.’

*While an impressive effort was made to identify and track motor units across a range of contractions, it appears that a substantial portion of muscle force was not identified. Though high-intensity contractions are challenging to decompose - the authors are commended for their technical ability to record population motor unit discharge times with recruitment thresholds up to 75% of a participant's maximal voluntary contractions. However previous groups have seen substantial recruitment of motor units above 80% and even 90% maximum activation in the soleus. Given the innervation ratios of higher threshold motor units, if recruitment continued to 100%, the top quartile would likely represent a substantial portion of the traditional fast-fatigable motor units. It would be highly interesting to understand the recruitment and rate coding of the highest threshold motor units, at a minimum I would suggest using terms other than "entire range" or "full spectrum of recruitment thresholds"*

Motor units were indeed identified between 0 and 80% of the maximal force in this study. This is due to the requirements of the decomposition algorithm that needs sustained and stable contraction to converge toward a set of separation vectors that generate sparse spike trains. Thus, it was not possible for our participants to sustain contractions above 80%MVC without generating fatigue.

However, it is important to note that only a few motor units are recruited above 80% of the maximal force in the TA (Van Cutsem et al., 1998, J Physiol), as well as in other muscles of the lower limb (Oya et al., 2009, J Physiol; Aeles et al., 2020, J Neurophysiol). Thus, we may have

only missed a few motor units recruited above 80% of the maximal force. Nevertheless, we removed the terms ‘full spectrum of recruitment thresholds’ and ‘entire range’ from the manuscript to now read ‘most of the spectrum of recruitment thresholds observed in humans.’.

*The quantification of hysteresis using torque appears to make self-evident the observation that lower threshold motor units demonstrate less hysteresis with respect to torque. If there is motor unit discharge there will be force. I believe this limitation goes beyond the floor effects discussed in the manuscript. Traditionally, individuals have used the discharge of a lower threshold unit as the measure on which to apply hysteresis analyses to infer ion channel function in human spinal motoneurons.*

We agree with the reviewer that the hysteresis is classically estimated using the firing rate of a ‘reporter unit’ with the delta F method (introduced in humans by Gorassini et al.), or most recently with the advances in motor unit identification using the cumulative spike train of the identified motor unit. The researchers use this data as a proxy of the synaptic drive, and compare their values at recruitment and derecruitment thresholds of the ‘test unit’.

As mentioned above in response to reviewer 1, this approach was not possible in our study as we did not have the same units across contractions to estimate cumulative spike trains. It was therefore not possible to pool the data across contractions as we did here to generate force/firing rate relations on the widest range of force. This limitation is now highlighted in the discussion section (P19; L470): ‘This result must be confirmed with a more direct proxy of the net synaptic drive, such as the firing rate of a reference low-threshold motor neuron used in the delta F method (Gorassini et al., 1998), or the cumulative spike train of low-threshold motor neurons (Afsharipour et al., 2020).’.

*The main findings are not entirely novel. See Monster and Chan 1977 and Kanosue et al 1979.*

We agree with the reviewer that the results of the paper are remarkably aligned with previous experimental findings in humans, in animals, or with in vitro and in silico models. However, we believe that our study shows in humans the incredible variety of rate coding patterns within a pool of motor units that span most of the spectrum of recruitment thresholds observed in humans. It also highlights the variability of rate coding patterns between motor neurons that have a similar recruitment threshold. Finally, we observe differences between pools of motor neurons innervating two different muscles in the lower limb, mirroring what has been done in the past in the upper limb muscle.

#### **Recommendations for the authors:**

##### **Reviewer #1 (Recommendations For The Authors):**

*The wording 'decode' across the manuscript may sound somewhat unsuitable for the context, because 'decode' would involve interpreting the signals and activities to understand how they relate to specific variables or proxies of behavior. Here in this study it does not necessarily involve the interpretation, but sounds to be used for decomposing the signal into the constituent motor units. As such, it might be appropriate to use other words such as decompose, read out, or extract.*

‘Decode’ was removed from the manuscript to now read motor unit ‘identification’

**Reviewer #2 (Recommendations For The Authors):**

*Figures 1 and 2 are informative and interesting. Figures 3 and 4 are harder to interpret. For example, in Figure 4, data plotted along the diagonal is overplotted and not as informative.*

For the sake of clarity, we separated the lines of the fits and the scatter plots in the right panels in Figure 3. In Figure 4, we remove the scatter plots and only reported the lines of the fits for each participant.

*Do you think the different durations of the isometric plateau across contraction intensities influenced motor unit derecruitment? Longer duration in lower threshold motor units would have resulted in a larger effect of PICs?*

We did not find an effect of the duration of the plateau on the derecruitment threshold. Notably, a computational study found that the duration of the plateau may impact the delta F, due to the combination of PICs, spike threshold accommodation and spike frequency adaptation (Revill & Fuglevand, 2011, J Neurophysiol). However, we did not use the delta F value here to estimate the effect of PICs on the hysteresis.

*L703. For the measure of firing rate hysteresis the difference between recruitment and derecruitment was calculated, but why not use the delta-F method? This is more commonly used to assess hysteresis as a rough estimate of intrinsic dynamics.*

As further discussed above, this approach was not possible in our study as we did not have the same units across contractions to estimate cumulative spike trains. It was therefore not possible to pool the data across contractions as we did here to generate force/firing rate relations on the widest range of force.

This was mentioned in the discussion (P19; L470):

‘This result must be confirmed with a more direct proxy of the net synaptic drive, such as the firing rate of a reference low-threshold motor neuron used in the delta F method (Gorassini et al., 1998), or the cumulative spike train of low-threshold motor neurons (Afsharipour et al., 2020).’

*L144. The standard deviation seems high. Some participants had fewer than 20 motor units and your number of participants per muscle was eight, could you state the complete range?*

A table was added in the results section to indicate the yields of the decomposition per contraction.

*If other studies are able to randomly sample motor units with intramuscular electrodes does this also represent an estimate of rate coding from the 'entire' pool? One criticism of HDsEMG arrays is that they are biased towards decomposing superficial larger motor units and in the male sex.*

The decomposition of EMG signals recorded with arrays of surface electrodes is indeed biased toward the identification of motor units with the larger action potentials in the signal (large and superficial; Farina & Holobar, 2016, Proceedings of IEEE). We took advantage of the latter limitation by performing successive contractions at different levels of force with the objective to identify the last recruited motor units (larger units according to the size principle), while tracking the smaller ones. In that way, we were able to sequentially identify motor units

recruited from 0% to 75% of the maximal force. A similar approach could be applied to selective intramuscular electrodes. However, because identifying motor units up to maximal force requires a highly selective pair of fine wires or needle electrodes, the procedure described above should be repeated hundreds of times to reach the same samples as those obtained in our study.

*L151-161. The ratio between simulated and decomposed surface EMG reached 55% for the TA and 70% for the VL. How does this provide support that the "entire" MU pool was sampled?*

As said above, we do not identify all the motor units during each contraction, but rather the larger ones with the larger action potentials within the EMG signals. However, we used here a sequential approach to identify new motor units during each trial while tracking smaller units. In that way, we were able to sequentially identify on average 130 motor units per muscle.

To avoid any confusion, we removed the references to 'entire' pools in the manuscript.

*L266. How is it possible that in some participants no motor units were recruited below 5% of MVC? Do the authors suspect they produced force from synergist muscles or that the decomposition failed to identify these presumably smaller and deeper motor units?*

This mostly results from the limitations of the decomposition algorithm. In these participants, it is likely that the decomposition was biased toward motor units only active during the plateau of force or recruited at the end of the ramp.

*Figure 2B. Do the higher threshold motor units with linear responses receive more inhibitory input (coactivation) or are devoid of large PIC effects?*

*Were antagonist muscles recorded? During higher contraction intensities, greater antagonist coactivation in some trials or participants may have linearized the firing rate profiles (e.g., Revill and Fuglevand, 2017).*

*L427. This is a neat finding that higher threshold motor units are less likely to have the functional hallmark of a strong PIC effect and may therefore be more representative of extrinsic inputs. Could this be an advantage to increase the precision of stronger contractions or reduce the fatigability of muscle fibres during repeated strong contractions?*

Synaptic contacts with Renshaw cells (Fyffe, 1991, J Neurophysiol) and Ia inhibitory interneurons (Heckman & Binder, 1991, J Neurophysiol) are widespread within pools of motor units, which induces homogeneously distributed inhibitory inputs. However, the amplitude of these inhibitory inputs can increase with muscle force. We found that the EMG amplitude of the soleus and the gastrocnemius medialis recorded with bipolar EMG during the dorsiflexion increased with the force. Therefore, the higher inhibitory at higher force may also contribute to the linearisation of the force/firing rate relations observed with high threshold motor neurons, as suggested by Revill and Fuglevand (2017, J Physiol).

We discussed this point in the new version of the manuscript (P17; L415):

'The level of recurrent and reciprocal inhibition has also probably increased with the increase in force during the ramp up, progressively blunting the effect of persistent inward currents for late-recruited motor units (Kuo et al., 2003; Hyngstrom et al., 2007; Revill and Fuglevand, 2017). This may also explain the larger percentage of high-threshold motor units with a linear fit for the firing rate/force relation (Figure 2), as the integration of larger inhibitory inputs should linearise the firing rate/force relation (Revill and Fuglevand, 2017).'

*In Figure 2B, it makes sense that linear firing rate responses occur later in the ramp contraction when myotendinous slack is lower. Do the authors think contractile dynamics are matched to the firing rate profiles?*

To our knowledge, there is no direct data on the link between the linearity of the force/firing rate relation and the stiffness of the tendon. A recent work from Mazzo et al. (2021, J Physiol) has shown that repeated stretches of calf muscles, which induce a decrease in their stiffness, induced an increase in motor unit firing rate at low levels of forces. This indicates that the contractile properties of the muscle may potentially also impact the profile of rate coding when considered as function of force.

We added this point in the discussion (P20; L512):

‘On a different note, the steep increase in firing rate over the first percentages of the ramp-up may also enable the motor units to produce the required level of force despite having a more compliant muscletendon unit (Mazzo et al., 2021).’

*L371. It is likely that Marshall et al., 2022, recorded over 100 unique motor units from the same animal.*

The reviewer is right that Marshall may have identified hundreds of motor units across sessions in one non-human primate. However, there is no ways to verify this statement as they used fine wire electrodes inserted in different locations in each session, which made it impossible to verify the uniqueness of each identified unit. Conversely, we verified in our study that all the motor units were unique using the distribution of their surface action potentials across the 236 surface electrodes.

*L378. What do the authors mean by "rate coding is similar"? I find this statement confusing. Is this regarding the absolute firing rate range, response to force increases, hysteresis, or how they scale with contraction intensity?*

This statement was removed from the discussion to avoid any confusion.

### **Reviewer #3 (Recommendations For The Authors):**

*The authors may want to consider other mechanisms of the linearization of discharge rates of medium and high threshold motor units. Monica's work may suggest that, over time, there is a subthreshold activation of the PIC, which serves to linearize the eventual suprathreshold activation underlying repetitive discharge. Additionally, Andy has shown that inhibitory drive from cutaneous inputs can linearize the initial acceleration of low threshold motor units - cutaneous inputs, or even Ib inputs, may be greater later in the contraction and serve to linearize discharge rates.*

We thank the reviewer for their input on the discussion, where we now discuss this point:

‘The level of recurrent and reciprocal inhibition has also probably increased with the increase in force during the ramp up, progressively blunting the effect of persistent inward currents for late-recruited motor units (Kuo et al., 2003; Hyngstrom et al., 2007; Revill and Fuglevand, 2017). This may also explain the larger percentage of high-threshold motor units with a linear fit for the firing rate/force relation (Figure 2), as the integration of larger inhibitory inputs should linearise the firing rate/force relation (Revill and Fuglevand, 2017).’.

*Lines 433 - intrinsic properties, in particular the afterhyperpolarization, will likely influence maximal discharge rate and provide a ceiling to the change in firing rate.*



This point is now discussed in the draft (P17; L428):

‘This difference may be explained by smaller excitatory synaptic inputs onto low- than high-threshold motoneurons (Powers and Binder, 2001; Heckman and Enoka, 2012), lower synaptic driving potential of the dendritic membrane (Powers and Binder, 2000; Cushing et al., 2005; Fuglevand et al., 2015), and longer and larger afterhyperpolarisation phase in low- than high-threshold motoneurons (Bakels and Kernell, 1993; Gardiner, 1993; Deardorff et al., 2013; Caillet et al., 2022).’

*The actual yield per contraction is not entirely clear. Figure S2 is quite nice in this regard, but a table with this and other information on it may be helpful. This would help with the beginning of the abstract and discussion when it is stated that, on average over 100 motor units were identified per person.*

We added a table in the results to give the number of motor units identified per contraction.

*Are the thin film units represented in S2 and S3?*

Only motor units identified from signals recorded with arrays of surface electrodes are presented in figures S2 and S3.

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